

USER INSTRUCTIONS:

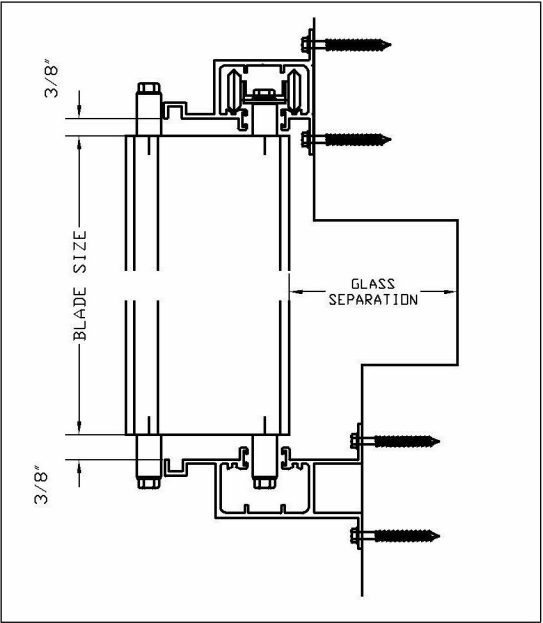
- 1- DETERMINE PROJECT DESIGN WIND LOAD PRESSURES AS PER ITEM 4 OF GENERAL NOTES, SHEET 1.
- 2- CHECK BLADE SIZE WITH TABLE 1 ON SHEET 1 FOR ALLOWABLE DESIGN PRESSURE.
- 3- SELECT MOUNT TYPE TO BE USED AS PER SHEETS 2, 3, 4, 5 & 6. MOUNT TYPES ARE INTERCHANGEABLE.
- 4- CALCULATE ANCHOR SPACING FUNCTION OF BLADE SIZE AND PRESSURE FOR EACH SPECIFIC MOUNT TYPE AS PER SHEETS 2, 3, 4, 5 & 6.

GENERAL NOTES: (FLORIDA BUILDING CODE 2023 - 8th edition)

- 1- DEFINITION: THIS PRODUCT IS AN ACCORDION TYPE SHUTTER; DESIGNED, CONSTRUCTED AND ERECTED TO EASILY ENCLOSE AN AREA, PROVIDING PROTECTION FROM HURRICANE FORCE WINDS AND WIND BORNE DEBRIS (LARGE & SMALL MISSILE) WITHIN THE ALLOWABLE DESIGNED PRESSURES AND LIMITATIONS STATED IN THIS APPROVAL.
- 2- CODE: THIS PRODUCT HAS BEEN TESTED AND DESIGNED IN ACCORDANCE WITH THE FLORIDA BUILDING CODE UNDER TAS 201, 202 AND 203.
- 3- POSTING: A PERMANENT LEGIBLE DECAL SHALL BE PLACED AT BOTTOM OF THE LOCKING BAR OR CENTER MATE FACING THE EXTERIOR OR OUTSIDE AS PER FLORIDA BUILDING CODE.
- 4- LOADS: DESIGNED LOAD CALCULATED BASED ON THE ASCE 7 AND PROVIDED BY A PROFESSIONAL ARCHITECT OR ENGINEER FOR EACH SPECIFIC PROJECT. THE CALCULATED DESIGNED PRESSURE MUST NOT EXCEED THE ALLOWABLE PRESSURES FOR EACH SHUTTER COMPONENT TO BE USED.
- 5- MATERIAL: ALL EXTRUDED ALUMINUM SHAPES SHALL BE MADE OF 6063-T6 OR AS NOTED.
- 6- FASTENERS: ASSEMBLY SCREWS AND ANCHORS SHALL BE AS SPECIFIED IN THE CURRENT SET OF DRAWINGS. INSTALLATION AND LOADS AS PER THIS APPROVAL.
- 7- USE: IT SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR, ARCHITECT OR ENGINEER OF RECORD TO VERIFY THE FOLLOWING:
- 7.1- THE STABILITY OF THE STRUCTURE WHERE THE SHUTTER IS TO BE ATTACHED INSURING PROPER ANCHORAGE.
- 7.2- THE SITE SPECIFIC PROJECT CRITERIA, SUCH AS BUT NOT LIMITED TO, WIND LOADS, LOCAL CODE REQUIREMENTS, DESIGNED PRESSURES ETC.
- 7.3- THAT THIS APPROVAL IS ADEQUATE TO THE SPECIFIC PROJECT.
- 8- 33 1/3% INCREASE IN ALLOWABLE LOADS HAVE NOT BEEN USED IN THE DESIGN OF THE ANCHORS FOR THIS PRODUCT APPROVAL. HOWEVER, WOOD ANCHORS WITH 1.6 LOAD DURATION INCREASE FOR WIND LOAD HAVE BEEN USED.
- 9- OFFSET REQUIREMENTS: PRODUCT IS QUALIFIED AS "NON POROUS", THEREFORE GLAZING SEPARATION (GS) AS PER TABLE 1 IS NOT REQUIRED WITH EXCEPTION FOR THE FOLLOWING LOCATIONS:
- 9.1- IN ESSENTIAL FACILITIES IN ALL WIND ZONES.
- 9.2- IN WIND ZONE 4 AS DEFINED IN ASTM E 1996 WITH MODIFICATION BY THE FBC SECTION 1609.1.2.2 (LOCATIONS WITH BASIC WIND SPEED GREATER THAN 170 MPH), WHERE THE SPECIFYING AUTHORITY HAS SPECIFIED OPTIONAL ADDITIONAL PASS/FAIL CRITERIA IN ACCORDANCE WITH ASTM E 1996-17 APPENDIX X4.
- 9.3- IN HIGH VELOCITY HURRICANE ZONES (HVHZ), MIAMI DADE AND BROWARD COUNTIES.

TABLE 1
ALLOWABLE DESIGNED LOADS
X
MAXIMUM BLADE SIZE

BLADE (IN) INCHES	PRESSURE (PSF)	
	POSITIVE	NEGATIVE
60"	188	188
66"	170	170
72"	156	156
78"	144	144
84"	134	134
90"	125	125
96"	117	117
102"	110	110
108"	102	102
114"	92	92
120"	83	83
126"	75	75
132"	69	69
138"	59	63
144"	49	58
150"	42	52



GLASS SEPARATION: SEE GENERAL NOTE 9 FOR REQUIREMENTS ON SHUTTER OFFSET FROM GLAZING PANELS

GLASS SEPARATION		
INSTALLATION BELOW 30' ELEVATION	BLADE <=120" 3 1/2"	BLADE >120" 4 3/4"
INSTALLATION ABOVE 30' ELEVATION	BLADE <=120" 3"	BLADE >120" 3"

TYPICAL ELEVATION

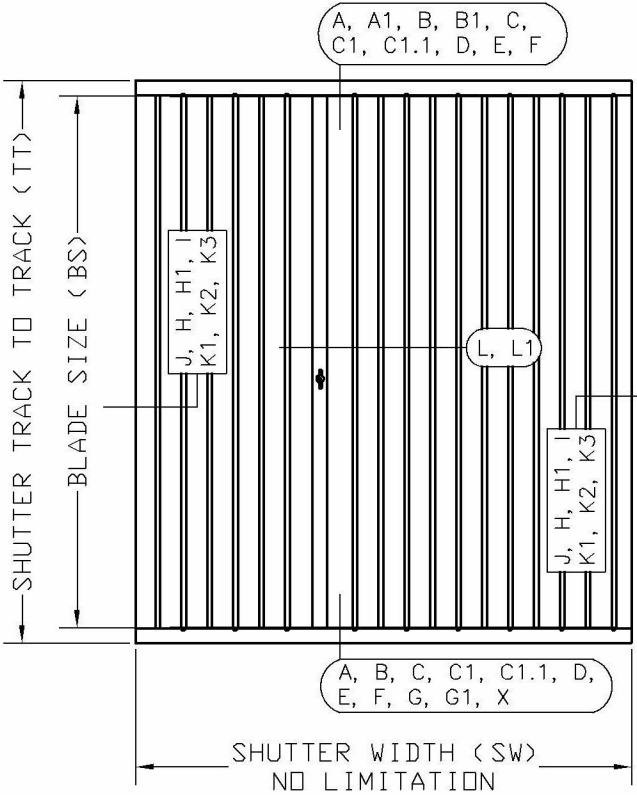


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SHEET 6- SECTIONS C & G I WITH ANCHORAGE TABLES, REMOVABLE TRACK
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Pedro De Figueiredo
PE 52609
09/10/2023

Pedro Figueiredo
Digitally signed by Pedro Figueiredo
Date: 2023.09.16 09:10:00 -04'00'

DIGITALLY SIGNED FOR 12 PAGES

THIS DOCUMENT HAS BEEN DIGITALLY SIGNED AND SEALED BY PEDRO DE FIGUEIREDO, PE 52609 ON THE DATE INDICATED HERE.

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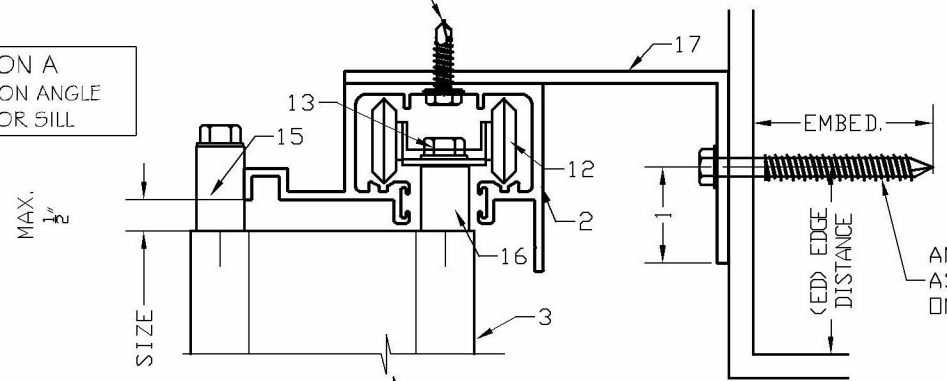
PRODUCT:
CROCI
ACCORDION SHUTTER
EXTRUDED ALUMINUM

MANUFACTURER:
CROCI NORTH AMERICA
11600 ADELMO LANE FORT MYERS, FL 33966
TEL.: (239) 278-3066 - FAX: (239) 278-5977
EMAIL: INFO@CROCIUSA.COM

SCALE:
NA
DATE:
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23-162
SHEET
1 of 12

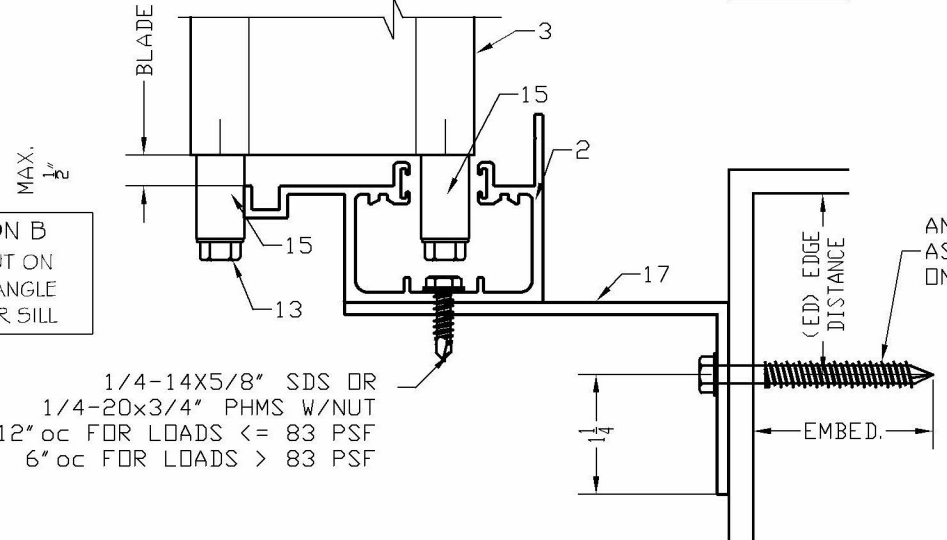
1/4-14X5/8" SDS OR
1/4-20X3/4" PHMS W/NUT
12" oc FOR LOADS <= 83 PSF
6" oc FOR LOADS > 83 PSF

SECTION A
BUILT-OUT ON ANGLE
HEADER OR SILL



ANCHORS DESIGNED
AS PER SECTION A
ON TABLE BELOW

SECTION B
BUILT-OUT ON
REVERSE ANGLE
HEADER OR SILL



ANCHORS DESIGNED
AS PER SECTION B
ON TABLE BELOW

1/4-14X5/8" SDS OR
1/4-20X3/4" PHMS W/NUT
12" oc FOR LOADS <= 83 PSF
6" oc FOR LOADS > 83 PSF

ALLOWABLE
PRESSURE
(PSF)
MAXIMUM
BLADE SPAN
IN INCHES

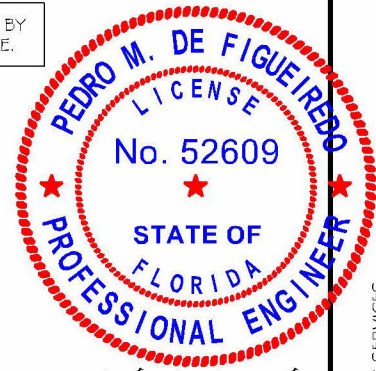
		MAXIMUM ANCHORAGE SPACING IN INCHES ON CENTER											
		SECTION A SILL / HEADER				SECTION B SILL / HEADER				SECTION C I SILL / HEADER			
BS	Pd	A	B	C	E	A	B	C	E	A	B	C	E
60	188	9	--	12	7	5	--	7	4	7	--	8	5
	140	12	4	12	10	9	--	11	7	12	--	12	9
	100	12	5	12	12	12	5	12	12	12	5	12	12
	60	12	9	12	12	12	9	12	12	12	9	12	12
	40	12	12	12	12	12	12	12	12	12	12	12	12
90	188	6	--	7	4	--	--	4	--	--	--	4	--
	160	7	--	8	5	4	--	5	3	4	--	5	3
	125	9	--	12	7	5	--	7	4	7	--	8	5
	80	12	4	12	12	12	--	12	9	12	4	12	12
	35	12	10	12	12	12	10	12	12	12	10	12	12
108	125	7	--	9	6	4	--	5	3	5	--	6	4
	100	10	--	12	8	6	--	7	4	7	--	9	5
	80	12	--	12	10	8	--	10	6	11	--	12	8
	70	12	4	12	11	11	--	12	8	12	4	12	11
	35	12	8	12	12	12	8	12	12	12	8	12	12
120	105	8	--	10	6	4	--	6	3	5	--	7	4
	90	10	--	12	8	6	--	7	4	7	--	9	5
	85	11	--	12	8	6	--	8	5	8	--	10	6
	60	12	4	12	12	12	--	12	9	12	4	12	12
	35	12	8	12	12	12	8	12	12	12	8	12	12
150	85	8	--	10	6	4	--	5	3	5	--	7	4
	65	12	--	12	9	7	--	8	5	9	--	11	6
	55	12	4	12	10	9	--	11	7	12	--	12	9
	45	12	5	12	12	12	4	12	10	12	5	12	12
	35	12	6	12	12	12	6	12	12	12	6	12	12

REFER TO ANCHOR SCHEDULE
ON SHEET 10 FOR ANCHOR TYPE

NOTES:
1- USE IN CONJUNCTION WITH TABLE 1, THE LOWER OF THE PRESSURE CONTROL DESIGN.
2- LINEAR INTERPOLATION IS ALLOWABLE BETWEEN BLADE SPAN SIZES

TYPICAL VERTICAL SECTIONS
AND ANCHOR SCHEDULE

THIS ITEM HAS BEEN DIGITALLY SIGNED AND SEALED BY
PEDRO DE FIGUEIREDO ON THE DATE INDICATED HERE.



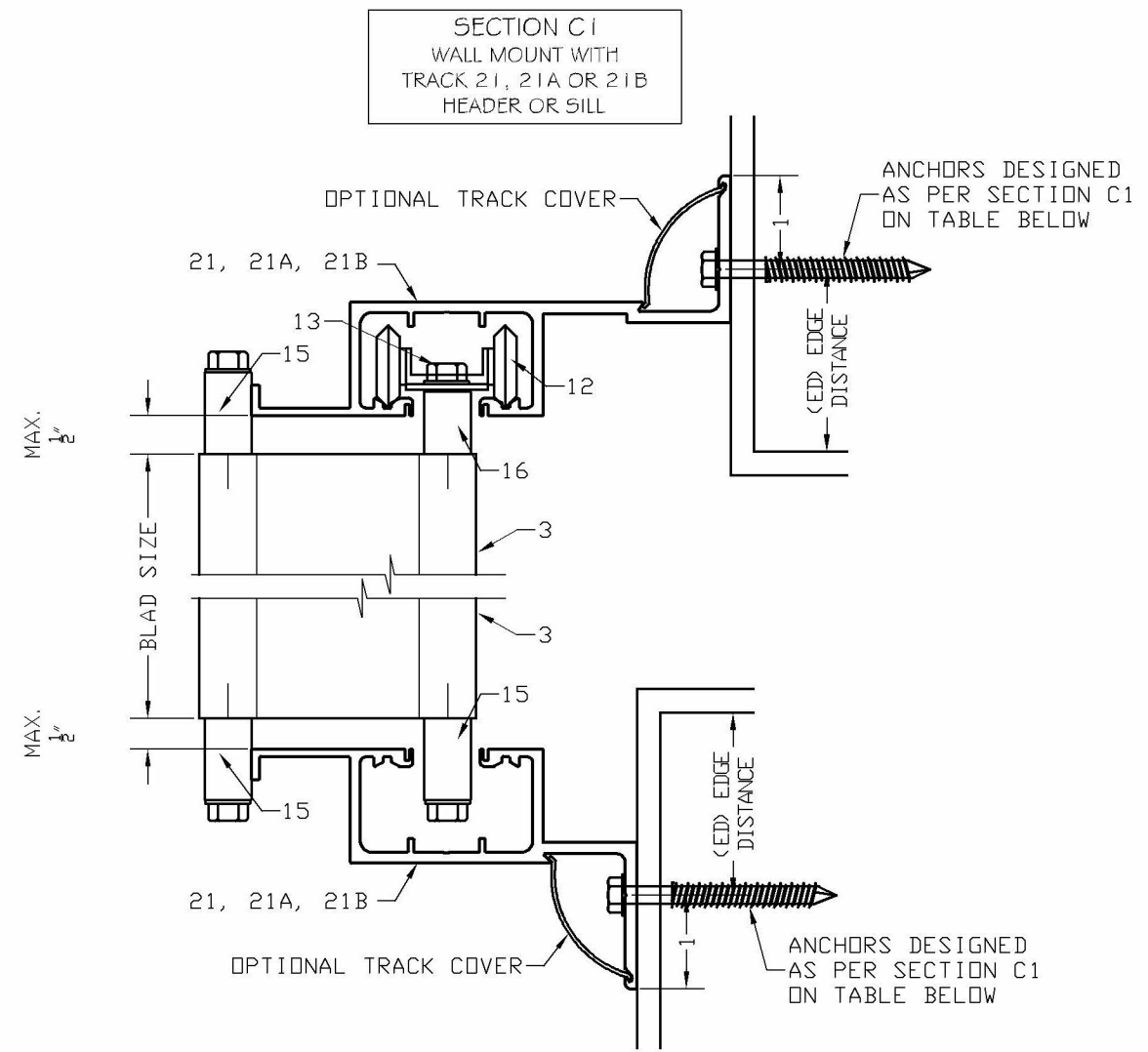
Pedro De Figueiredo
PE 52609
09/10/2023



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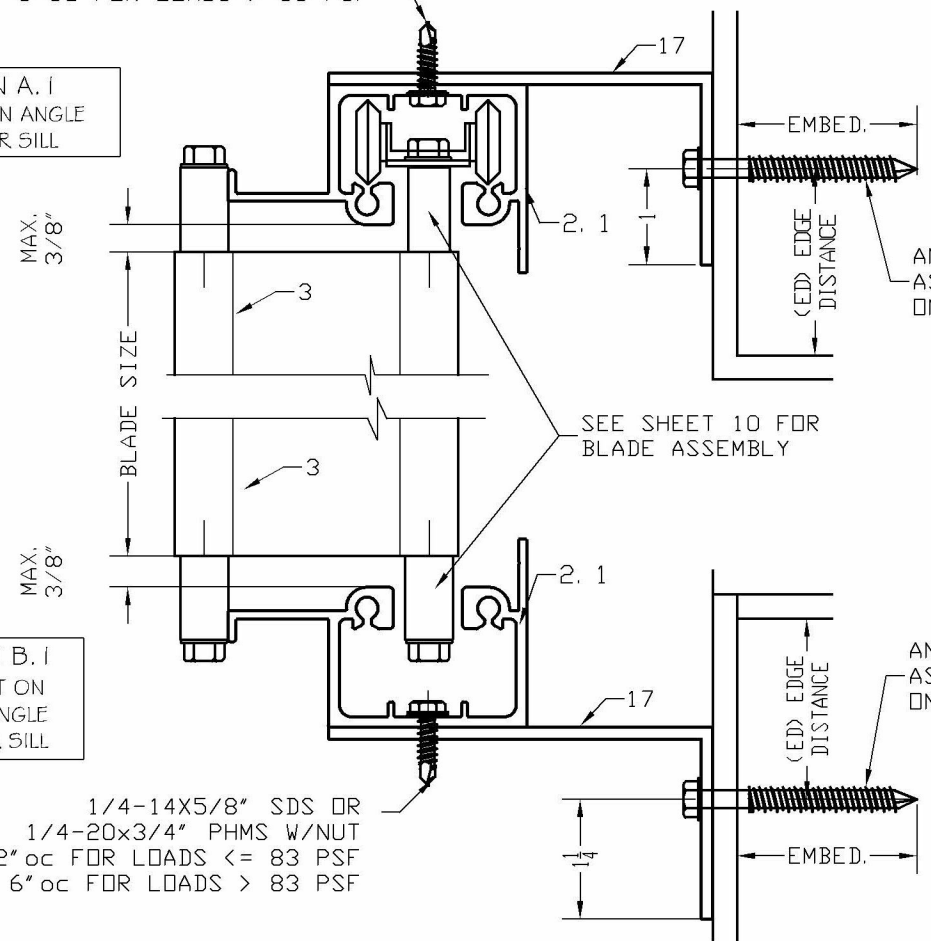


ANCHORS DESIGNED
AS PER SECTION C I
ON TABLE BELOW

ANCHORS DESIGNED
AS PER SECTION C I
ON TABLE BELOW

1/4-14X5/8" SDS OR
1/4-20X3/4" PHMS W/NUT
12" oc FOR LOADS <= 83 PSF
6" oc FOR LOADS > 83 PSF

SECTION A.1
BUILT-OUT ON ANGLE
HEADER OR SILL



SECTION B.1
BUILT-OUT ON
REVERSE ANGLE
HEADER OR SILL

1/4-14X5/8" SDS OR
1/4-20X3/4" PHMS W/NUT
12" oc FOR LOADS <= 83 PSF
6" oc FOR LOADS > 83 PSF

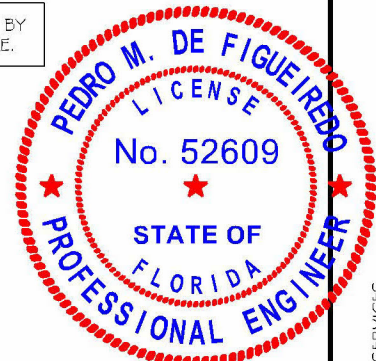
ALLOWABLE
PRESSURE
(PSF)
MAXIMUM
BLADE SPAN
IN INCHES

MAXIMUM ANCHORAGE SPACING IN INCHES ON CENTER														
SECTION A. I SILL / HEADER					SECTION B. I SILL / HEADER					SECTION C. I SILL / HEADER				
	A	B	C	E	A	B	C	E		A	B	C	E	
8	9	--	12	7	5	--	7	4		7	--	8	5	
0	12	4	12	10	9	--	11	7		12	--	12	9	
0	12	5	12	12	12	5	12	12		12	5	12	12	
	12	9	12	12	12	9	12	12		12	9	12	12	
	12	12	12	12	12	12	12	12		12	12	12	12	
8	6	--	7	4	--	--	4	--		--	--	4	--	
0	7	--	8	5	4	--	5	3		4	--	5	3	
6	9	--	12	7	5	--	7	4		7	--	8	5	
	12	4	12	12	12	--	12	9		12	4	12	12	
	12	10	12	12	12	10	12	12		12	10	12	12	
	7	--	9	6	4	--	5	3		5	--	6	4	
0	10	--	12	8	6	--	7	4		7	--	9	5	
	12	--	12	10	8	--	10	6		11	--	12	8	
	12	4	12	11	11	--	12	8		12	4	12	11	
	12	8	12	12	12	8	12	12		12	8	12	12	
6	8	--	10	6	4	--	6	3		5	--	7	4	
	10	--	12	8	6	--	7	4		7	--	9	5	
	11	--	12	8	6	--	8	5		8	--	10	6	
	12	4	12	12	12	--	12	9		12	4	12	12	
	12	8	12	12	12	8	12	12		12	8	12	12	
	8	--	10	6	4	--	5	3		5	--	7	4	
	12	--	12	9	7	--	8	5		9	--	11	6	
	12	4	12	10	9	--	11	7		12	--	12	9	
	12	5	12	12	12	4	12	10		12	5	12	12	
	12	6	12	12	12	6	12	12		12	6	12	12	

REFER TO ANCHOR SCHEDULE
ON SHEET 10 FOR ANCHOR TYPE

TYPICAL VERTICAL SECTIONS
AND ANCHOR SCHEDULE

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Pedro De Figueiredo
PE 52609
09/10/2023

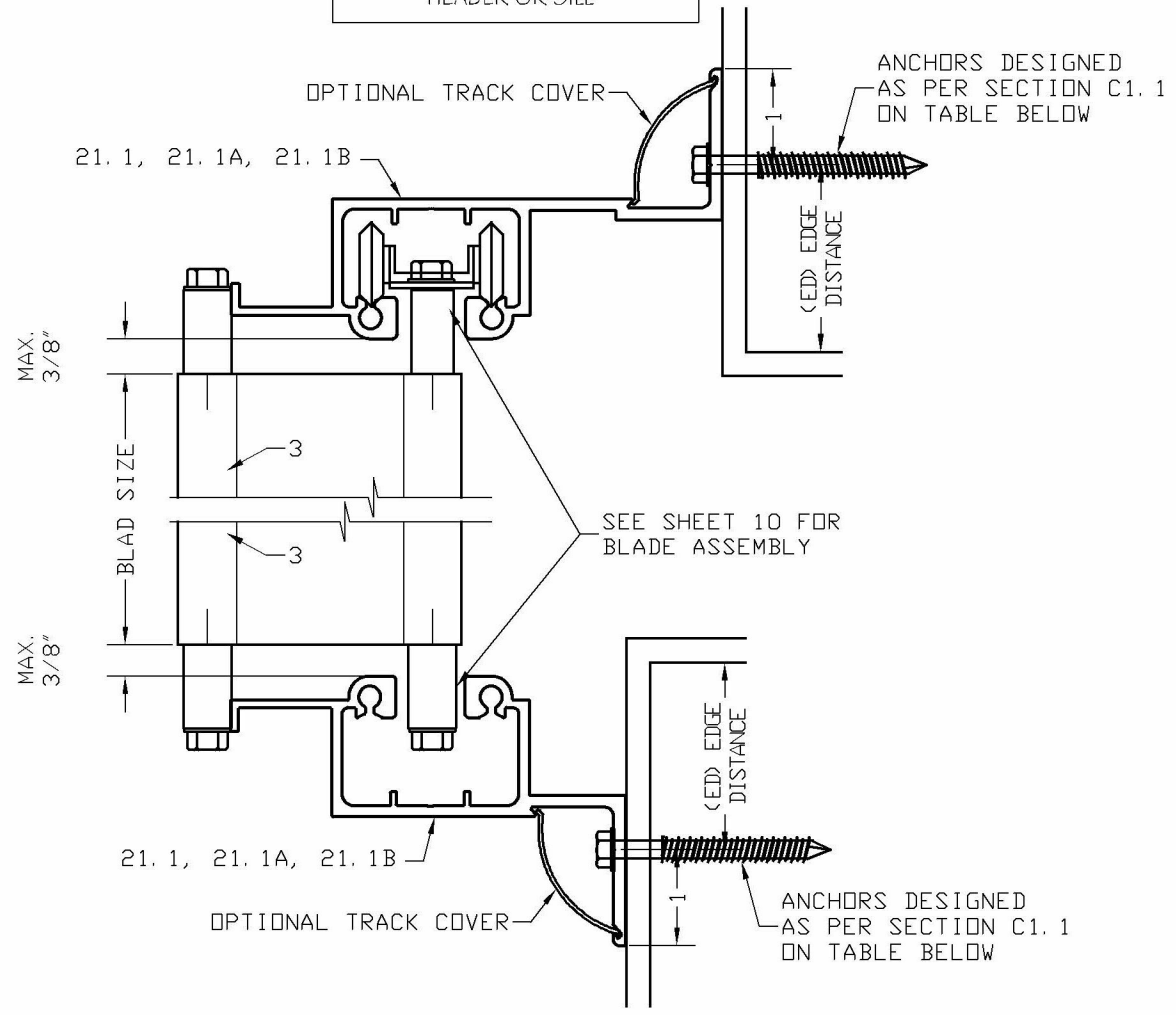


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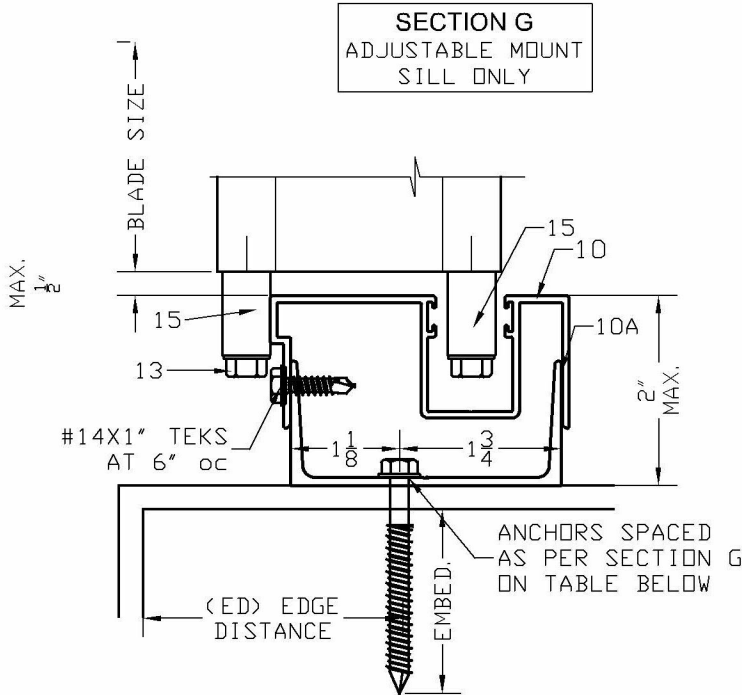
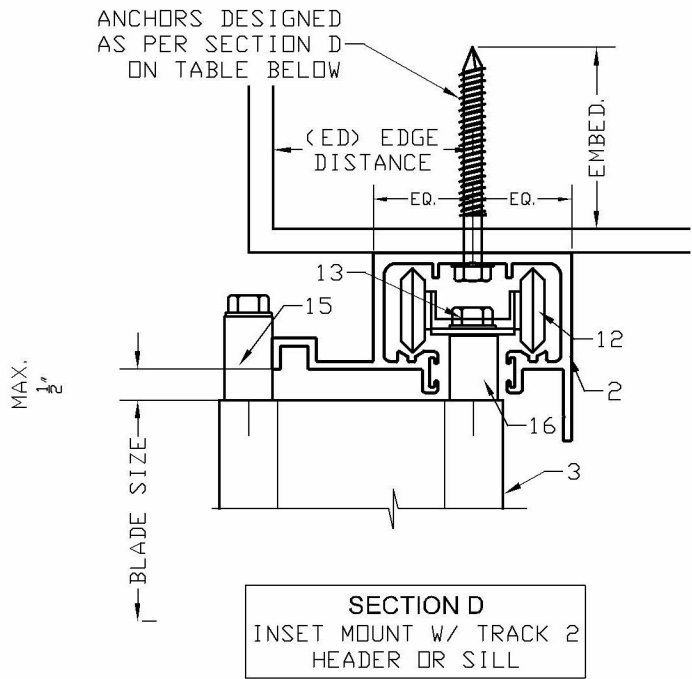
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SHEET
3 of 12

SECTION C1.1
WALL MOUNT WITH
TRACK 21.1, 21.1A OR 21.1B
HEADER OR SILL



NOTES:
1- USE IN CONJUNCTION WITH TABLE 1, THE LOWER OF THE PRESSURE CONTROL DESIGN.
2- LINEAR INTERPOLATION IS ALLOWABLE BETWEEN BLADE SPAN SIZES

TYPICAL VERTICAL SECTIONS
AND ANCHOR SCHEDULE



ALLOWABLE
PRESSURE
(PSF)

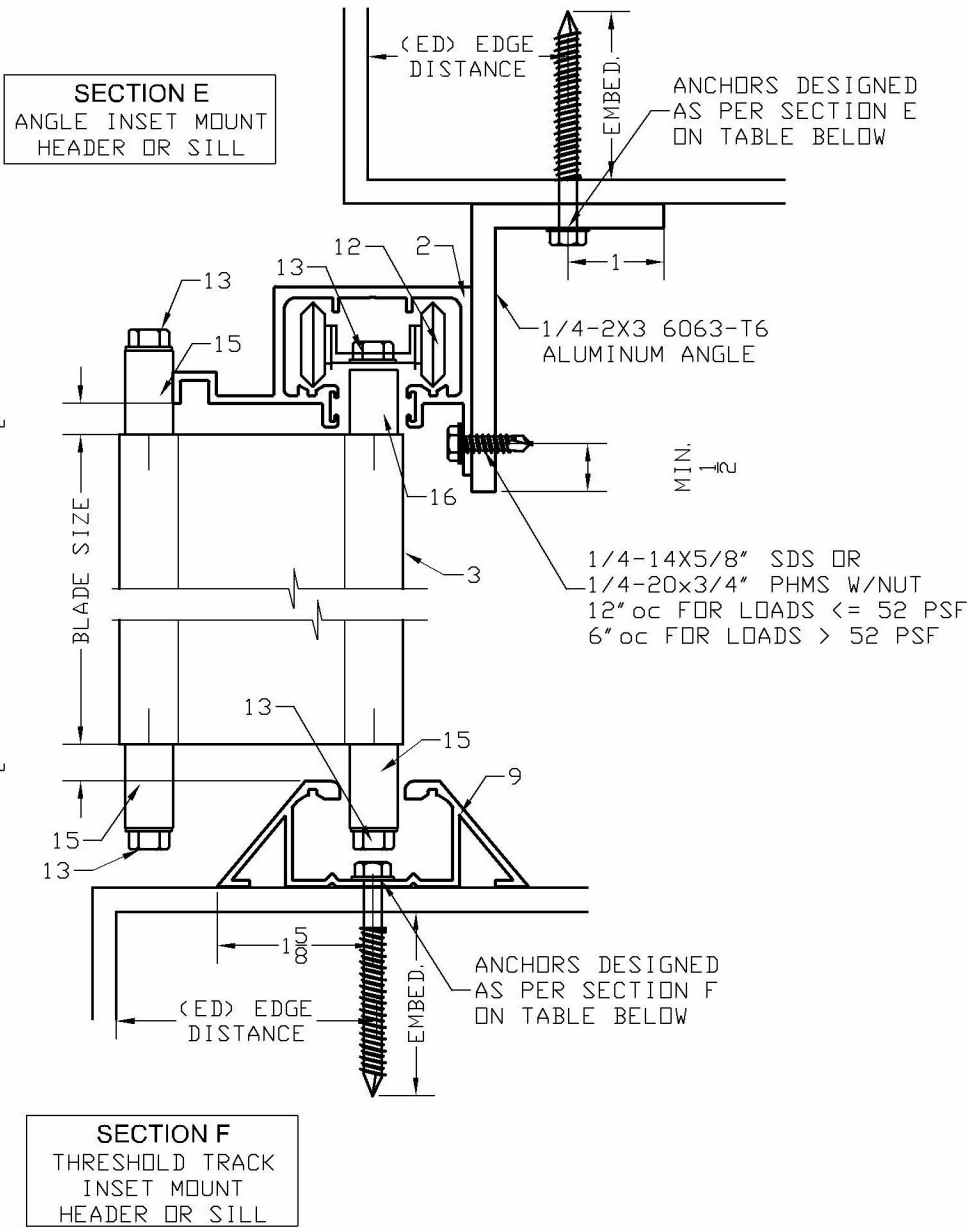
MAXIMUM
BLADE SPAN
IN INCHES

		MAXIMUM ANCHORAGE SPACING IN INCHES ON CENTER																	
		SECTION D SILL / HEADER				SECTION G SILL				SECTION E SILL / HEADER					SECTION F SILL / HEADER				
BS	Pa	A	B	C	E	A	B	C	E	A	B	C	D	E	A	B	C	E	
60	188	5	--	4	10	6	4	4	11	5	--	4	11	12	6	--	4	8	
	140	8	5	6	12	8	5	5	12	8	5	6	12	12	8	5	5	12	
	100	12	8	8	12	12	8	8	12	12	8	9	12	12	12	8	8	16	
	60	12	12	14	12	12	12	12	12	12	12	12	12	12	12	16	13	13	16
	40	12	12	12	12	12	12	12	12	12	12	12	12	12	12	16	16	16	16
90	188	--	--	--	4	--	--	--	5	--	--	--	5	4	--	--	--	4	
	160	4	--	--	6	4	--	3	8	--	--	3	6	6	4	--	3	5	
	125	5	--	4	10	6	4	4	11	5	--	4	11	12	6	--	4	8	
	80	10	6	7	12	10	6	6	12	10	6	7	12	12	10	6	6	14	
	35	12	12	12	12	12	12	12	12	12	12	12	12	12	16	15	15	16	
108	125	4	--	--	7	5	--	3	9	--	--	3	7	7	4	--	3	6	
	100	6	--	4	12	6	4	4	12	5	--	4	12	12	6	4	4	8	
	80	8	5	6	12	8	5	5	12	8	5	6	12	12	8	5	5	11	
	70	9	6	6	12	9	6	6	12	9	6	7	12	12	9	6	6	13	
	35	12	12	12	12	12	12	12	12	12	12	12	12	12	16	13	13	16	
120	105	4	--	--	8	5	--	3	10	4	--	3	8	8	5	--	3	6	
	90	6	--	4	12	6	4	4	12	5	--	4	12	12	6	4	4	8	
	85	6	4	4	12	7	4	4	12	6	4	5	12	12	6	4	4	9	
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	65	7	4	5	12	7	5	5	12	7	4	5	12	12	7	5	5	10	
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	45	10	7	7	12	10	7	7	12	10	7	8	12	12	10	7	7	14	
	35	12	9	9	12	12	9	9	12	12	9	10	12	12	13	9	9	16	

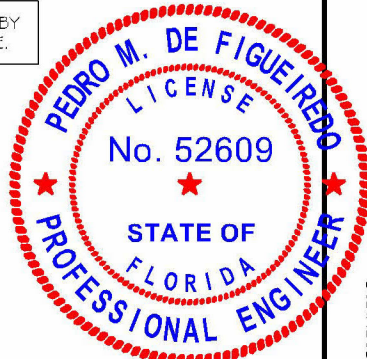
NOTES:

- 1- USE IN CONJUNCTION WITH TABLE 1, THE LOWER OF THE PRESSURE CONTROL DESIGN.
2- LINEAR INTERPOLATION IS ALLOWABLE BETWEEN BLADE SPAN SIZES

REFER TO ANCHOR
SCHEDULE ON SHEET
10 FOR ANCHOR TYPE



THIS ITEM HAS BEEN DIGITALLY SIGNED AND SEALED BY
PEDRO DE FIGUEIREDO ON THE DATE INDICATED HERE.



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DRAWING #

23-162

SHEET

4 of 12

ENGINEERING SERVICES

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PLANTATION, FL 33324

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ENGINEER SEAL

PEDRO DE FIGUEIREDO

PE 52609

SECTION X
FLOOR MOUNT
CONCEALED

SEE SHEET 10 FOR
BLADE ASSEMBLY

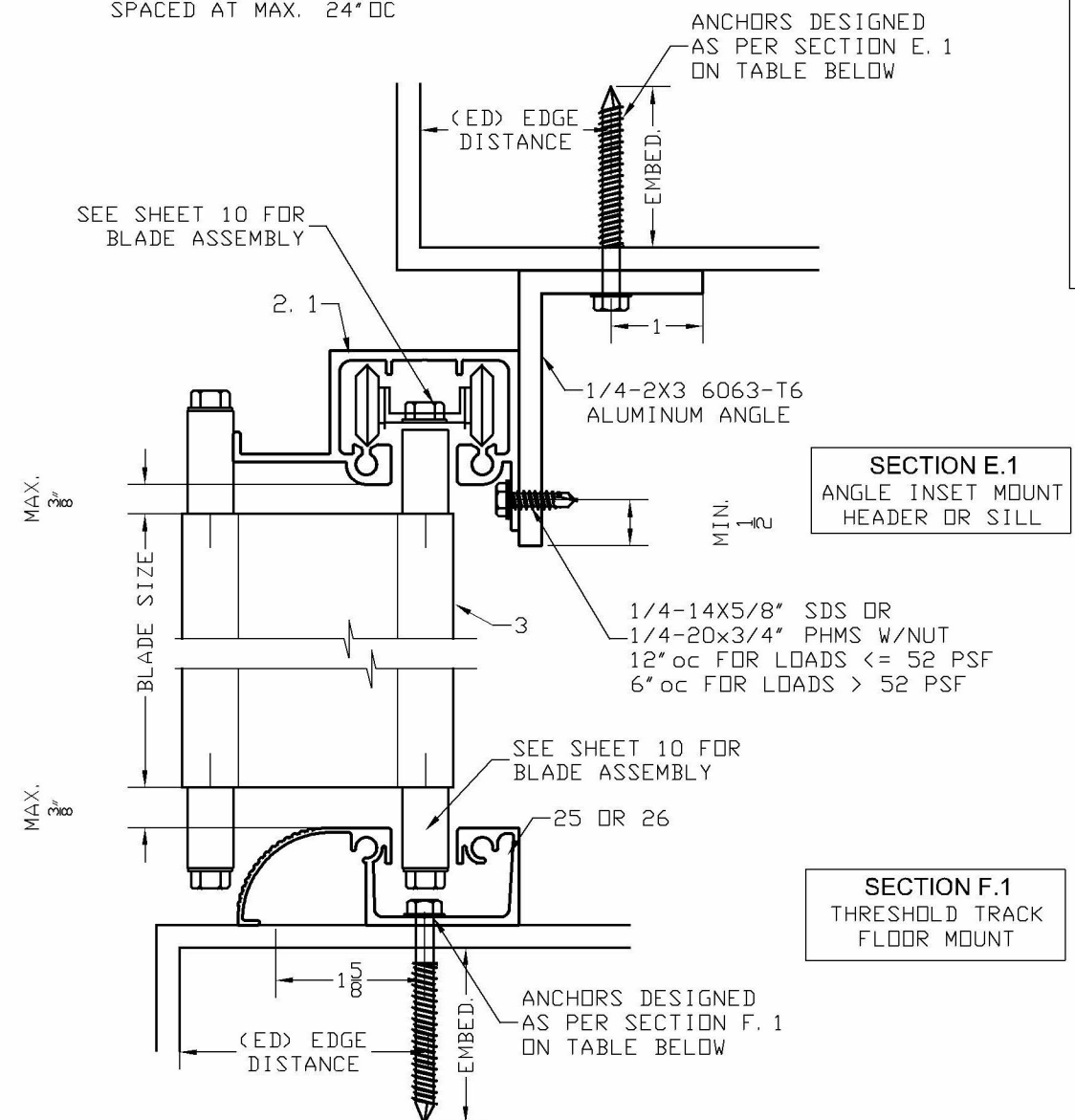
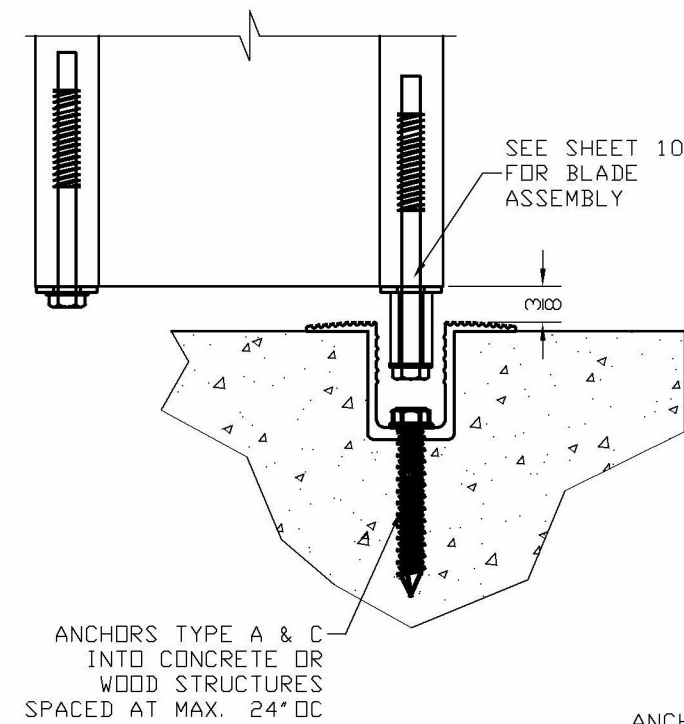
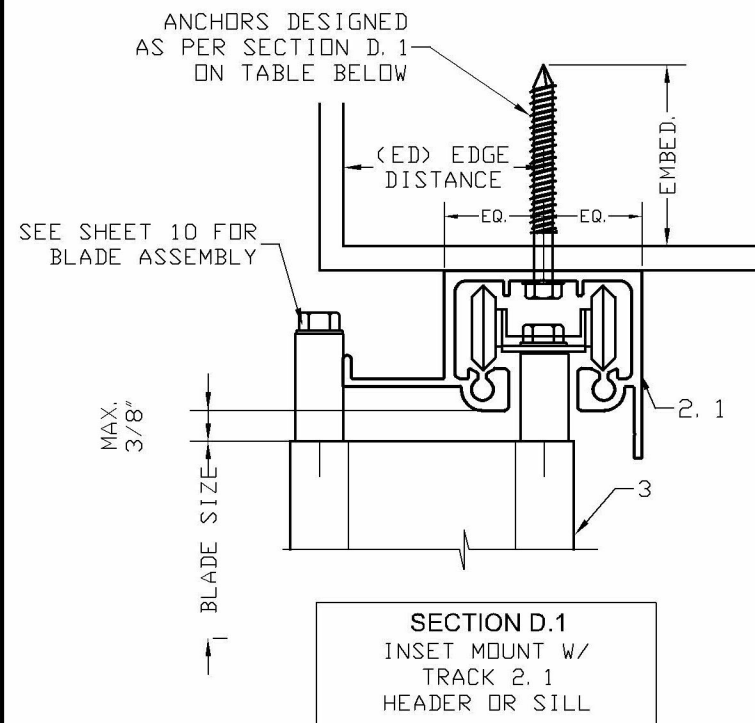
2-63

PAVERS ON SAND

2-25

2-100

ANCHORS TYPE A & C
INTO CONCRETE OR
WOOD STRUCTURES
PLACED AT MAX. 24" OC



ALLOWABLE PRESSURE (PSF) MAXIMUM BLADE SPAN IN INCHES		MAXIMUM ANCHORAGE SPACING IN INCHES ON CENTER															
		SECTION D. I SILL / HEADER				SECTION E. I SILL / HEADER					SECTION F. I SILL / HEADER						
BS	Pa	A	B	C	E	A	B	C	D	E	A	B	C	E			
60	188	5	--	4	10	5	--	4	11	12	6	--	4	8			
	140	8	5	6	12	8	5	6	12	12	8	5	5	12			
	100	12	8	8	12	12	8	9	12	12	12	8	8	16			
	60	12	12	14	12	12	12	12	12	12	16	13	13	16			
	40	12	12	12	12	12	12	12	12	12	16	16	16	16			
90	188	--	--	--	4	--	--	--	5	4	--	--	--	4			
	160	4	--	--	6	--	--	3	6	6	4	--	3	5			
	125	5	--	4	10	5	--	4	11	12	6	--	4	8			
	80	10	6	7	12	10	6	7	12	12	10	6	6	14			
	35	12	12	12	12	12	12	12	12	12	16	15	15	16			
108	125	4	--	--	7	--	--	3	7	7	4	--	3	6			
	100	6	--	4	12	5	--	4	12	12	6	4	4	8			
	80	8	5	6	12	8	5	6	12	12	8	5	5	11			
	70	9	6	6	12	9	6	7	12	12	9	6	6	13			
	35	12	12	12	12	12	12	12	12	12	16	13	13	16			
120	105	4	--	--	8	4	--	3	8	8	5	--	3	6			
	90	6	--	4	12	5	--	4	12	12	6	4	4	8			
	85	6	4	4	12	6	4	5	12	12	6	4	4	9			
	60	10	6	7	12	10	6	7	12	12	10	6	6	14			
	35	17	11	12	12	12	11	12	12	12	16	11	11	16			
150	85	4	--	--	8	4	--	3	8	8	5	--	3	6			
	65	7	4	5	12	7	4	5	12	12	7	5	5	10			
	55	8	5	6	12	8	5	6	12	12	8	5	6	12			
	45	10	7	7	12	10	7	8	12	12	10	7	7	14			
	35	12	9	9	12	12	9	10	12	12	13	9	9	16			

NOTES:
1- USE IN CONJUNCTION WITH TABLE 1, THE LOWER OF THE PRESSURE CONTROL DESIGN.
2- LINEAR INTERPOLATION IS ALLOWABLE BETWEEN BLADE SPAN SIZES



Pedro De Figueiredo
PE 52609
09/10/2023

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ENGINEERING SERVICES
CA 32459
9715 W. BOWARD BLVD. 247
PLANTATION, FL 33324
PPMF@AOL.COM

PRODUCT:
CROCI
ACCORDION SHUTTER
EXTRUDED ALUMINUM

MANUFACTURER:
CROCI
NORTH AMERICA
11600 ADELMO LANE FORT MYERS, FL 33966
TEL: (239) 278-3066 - FAX: (239) 278-5977
EMAIL: INFO@CROCIUSA.COM

SCAIF:

NA

DATE:
09/10/2023

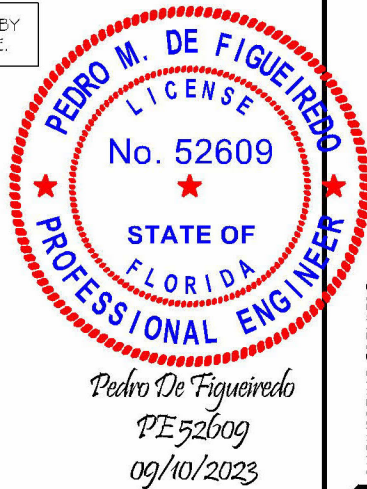
DRAWING #
23-162

SHEET
5 of 12

ALLOWABLE
PRESSURE
(PSF)
MAXIMUM
BLADE SPAN
IN INCHES

TYPICAL VERTICAL SECTIONS
AND ANCHOR SCHEDULE

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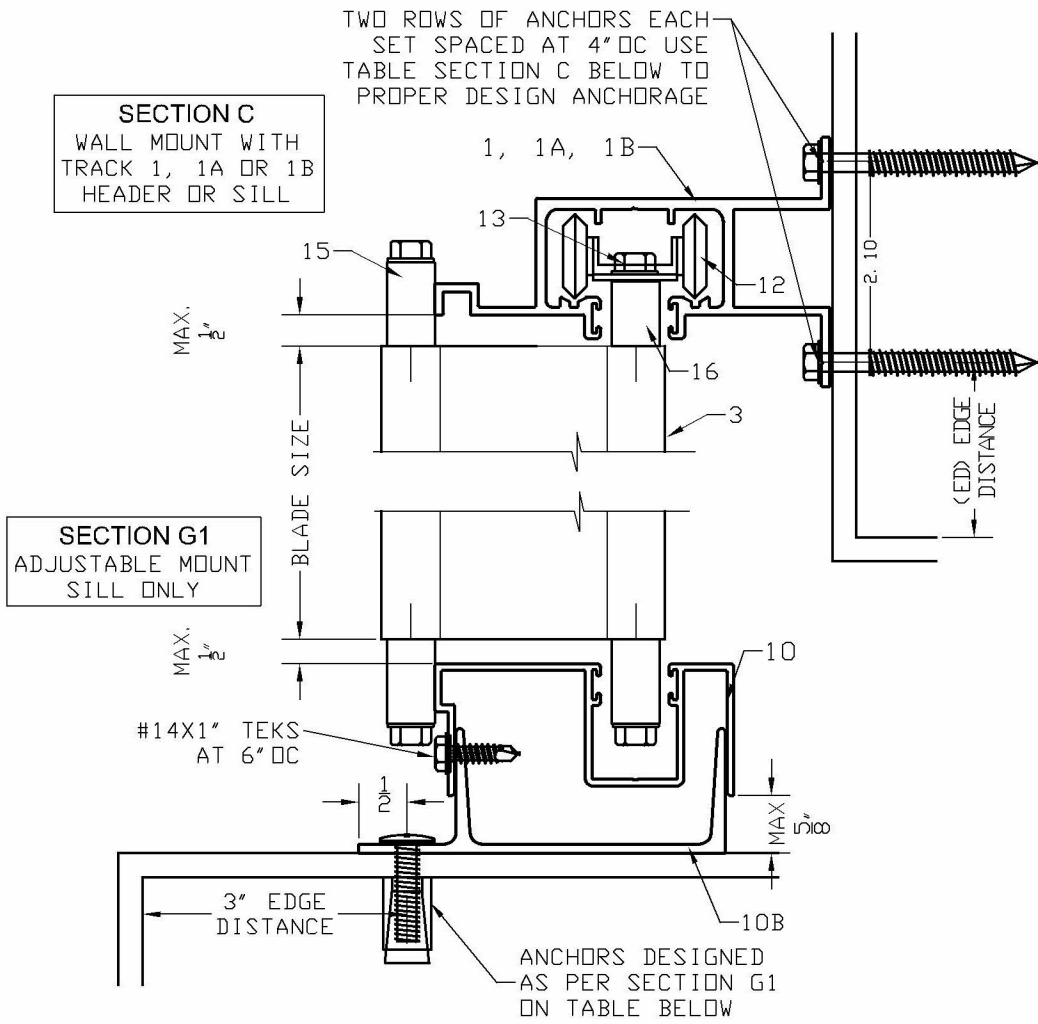
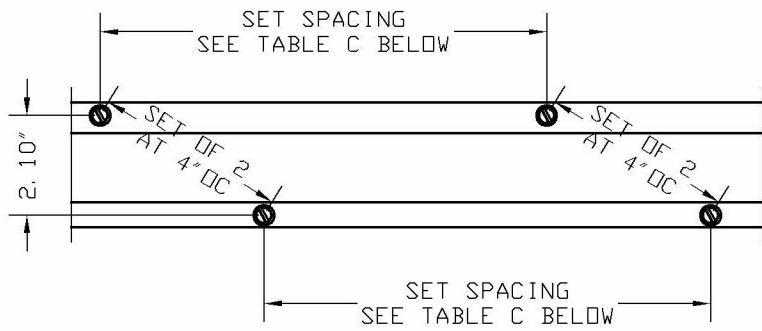
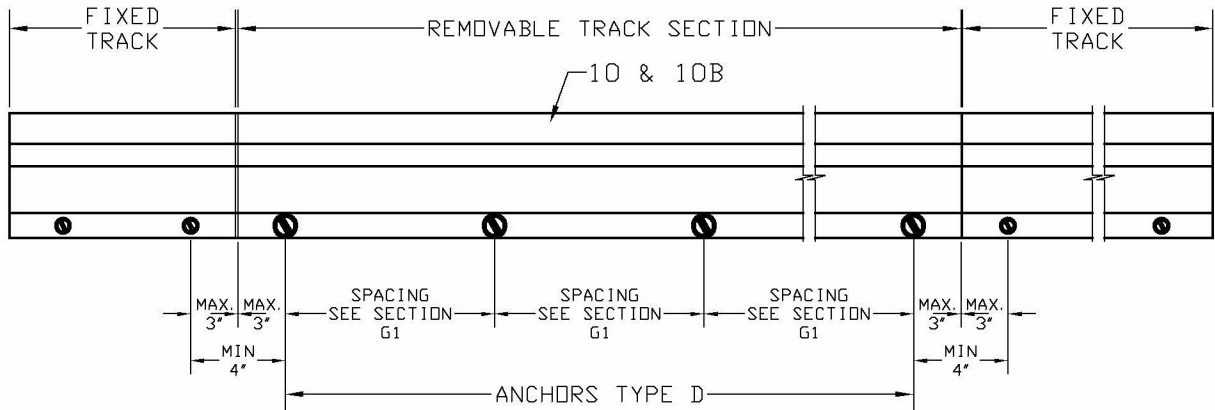
SCALE:
NA
DATE:
09/10/2023
DRAWING #
23-162
SHEET
6 of 12

		MAXIMUM ANCHORAGE SPACING IN INCHES ON CENTER									
		SECTION C SILL / HEADER					SECTION G1 SILL / HEADER				
BS	Pa	A	B	C	E	A	B	C	D	E	
60	188	22	6	24	17	6	4	4	12	11	
	140	24	8	24	21	8	5	5	12	12	
	100	24	11	24	24	12	8	8	12	12	
	60	24	18	24	24	12	12	12	12	12	
	40	24	24	24	24	12	12	12	12	12	
90	188	17	4	21	13	--	--	--	7	5	
	160	19	5	23	14	4	--	3	10	8	
	125	22	6	24	17	6	4	4	12	11	
	80	24	9	24	24	10	6	6	12	12	
	35	24	21	24	24	12	12	12	12	12	
108	125	20	5	24	15	5	--	3	12	9	
	100	22	6	24	17	6	4	4	12	12	
	80	24	7	24	20	8	5	5	12	12	
	70	24	8	24	23	9	6	6	12	12	
	35	24	17	24	24	12	12	12	12	12	
120	105	20	6	24	15	5	--	3	12	10	
	90	22	6	24	17	6	4	4	12	12	
	85	23	6	24	18	7	4	4	12	12	
	60	24	9	24	24	10	6	6	12	12	
	35	24	16	24	24	12	11	11	12	12	
150	85	20	5	24	15	5	--	3	12	10	
	65	24	6	24	18	7	5	5	12	12	
	55	24	8	24	21	8	5	6	12	12	
	45	24	10	24	24	10	7	7	12	12	
	35	24	12	24	24	12	9	9	12	12	

REFER TO ANCHOR
SCHEDULE ON SHEET
10 FOR ANCHOR TYPE

NOTES:
1- USE IN CONJUNCTION WITH TABLE 1, THE LOWER OF THE PRESSURE CONTROL DESIGN.
2- LINEAR INTERPOLATION IS ALLOWABLE BETWEEN BLADE SPAN SIZES

REMOVABLE SILL TRACK



TYPICAL HORIZONTAL SECTIONS
JAMB & CORNER SECTIONS

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09/10/2023



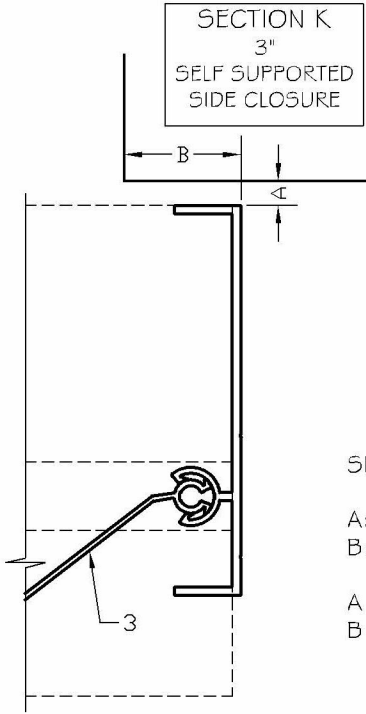
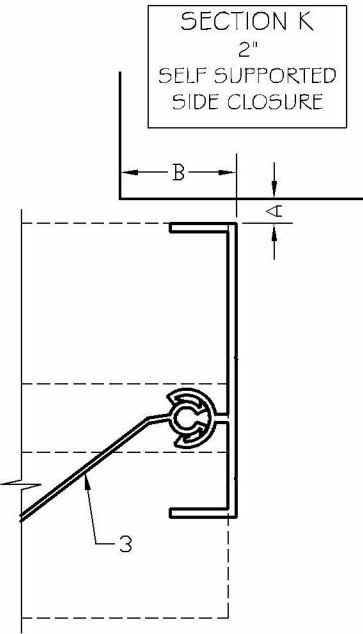
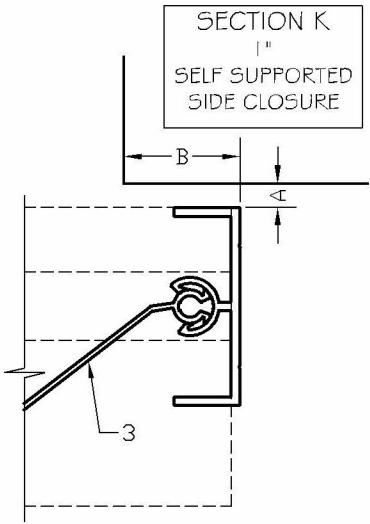
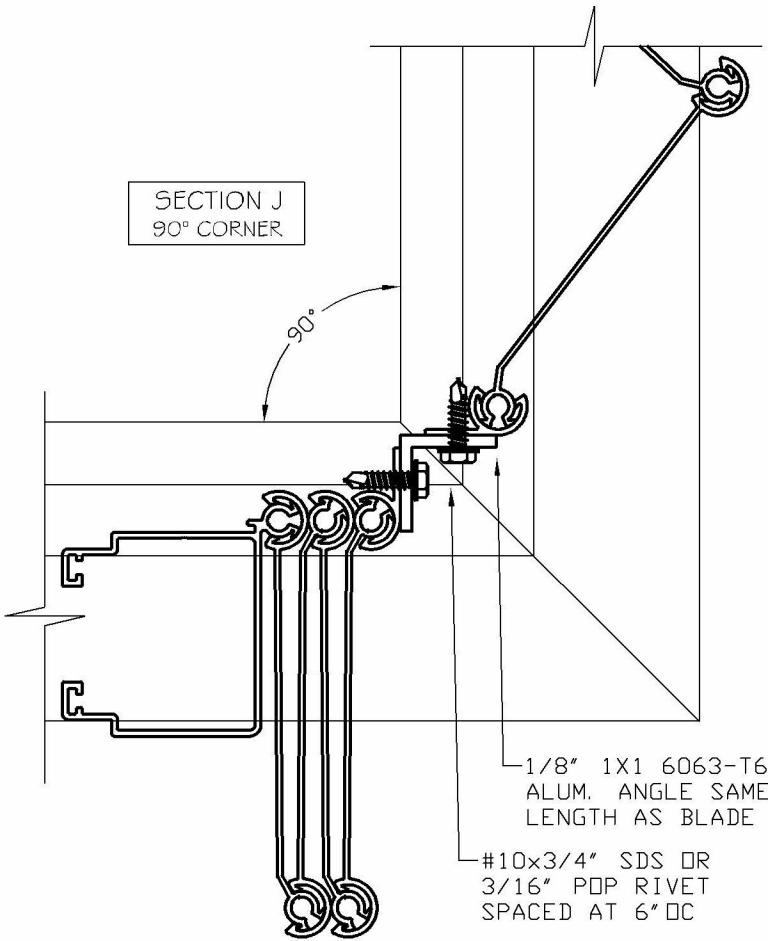
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CA 32459
9715 W. BROWARD BLVD. 247
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PPMIF@AOL.COM

ENGINEER SEAL
PEDRO DE FIGUEIREDO
PE 52609

PRODUCT:
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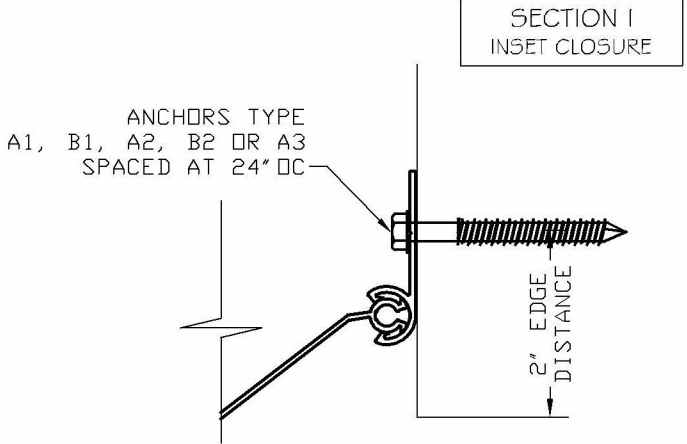
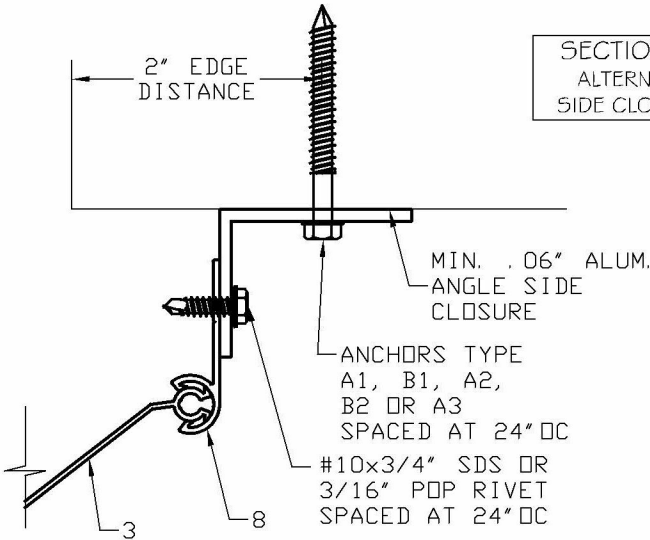
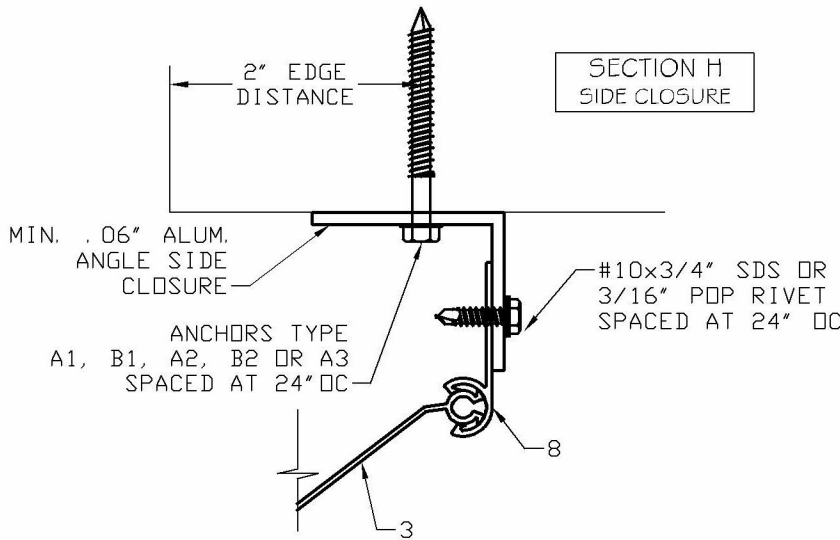
SCALE:
NA
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09/10/2023
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SECTION K - OPENING CLOSURE NOTES:

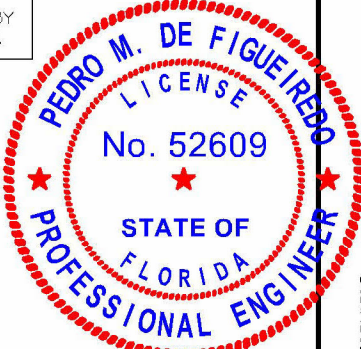
A: SIDE CLEARANCE
B: SHUTTER OVERLAP

A MUST BE NOT MORE THAN 1/4 INCHES OR
B GREATER THAN 1.5 X A



TYPICAL HORIZONTAL SECTIONS
LOCK SECTIONS

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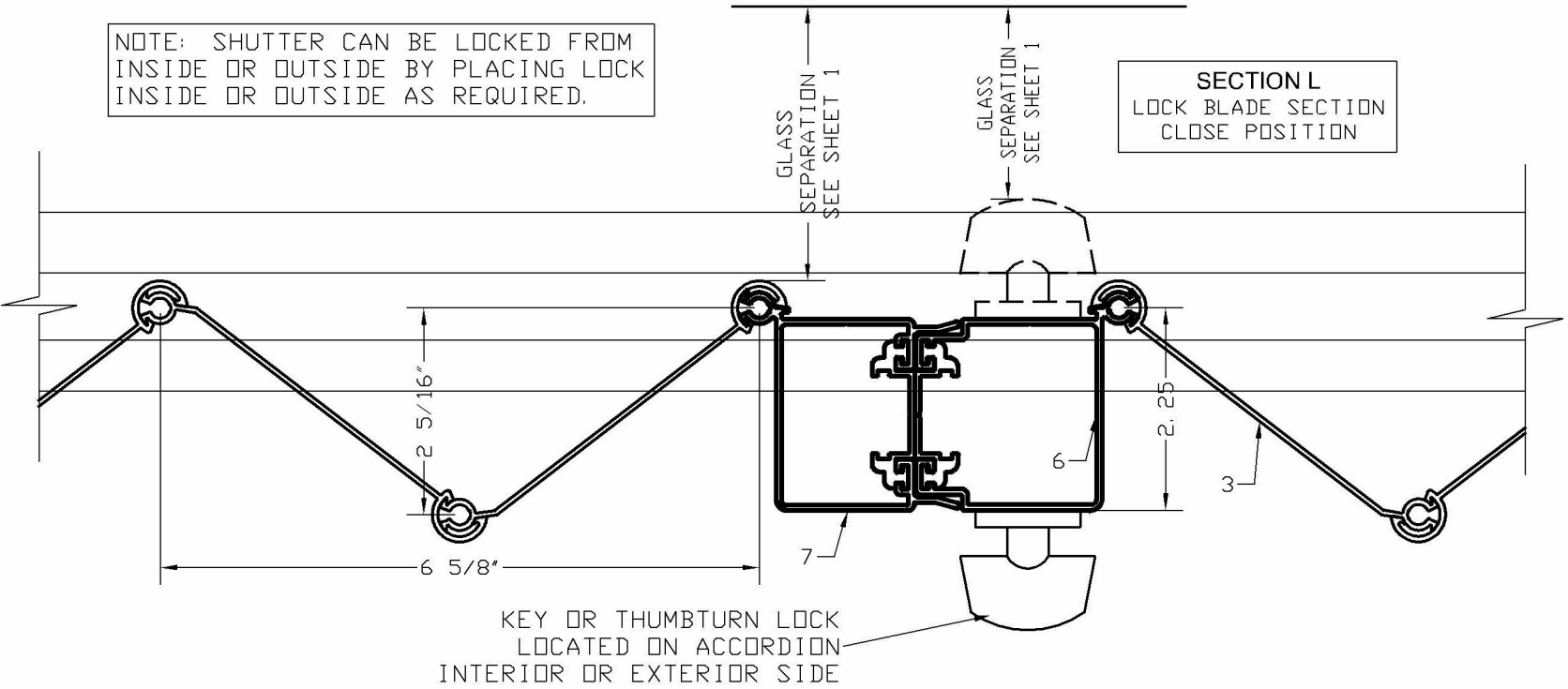
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PPMIF@AOL.COM

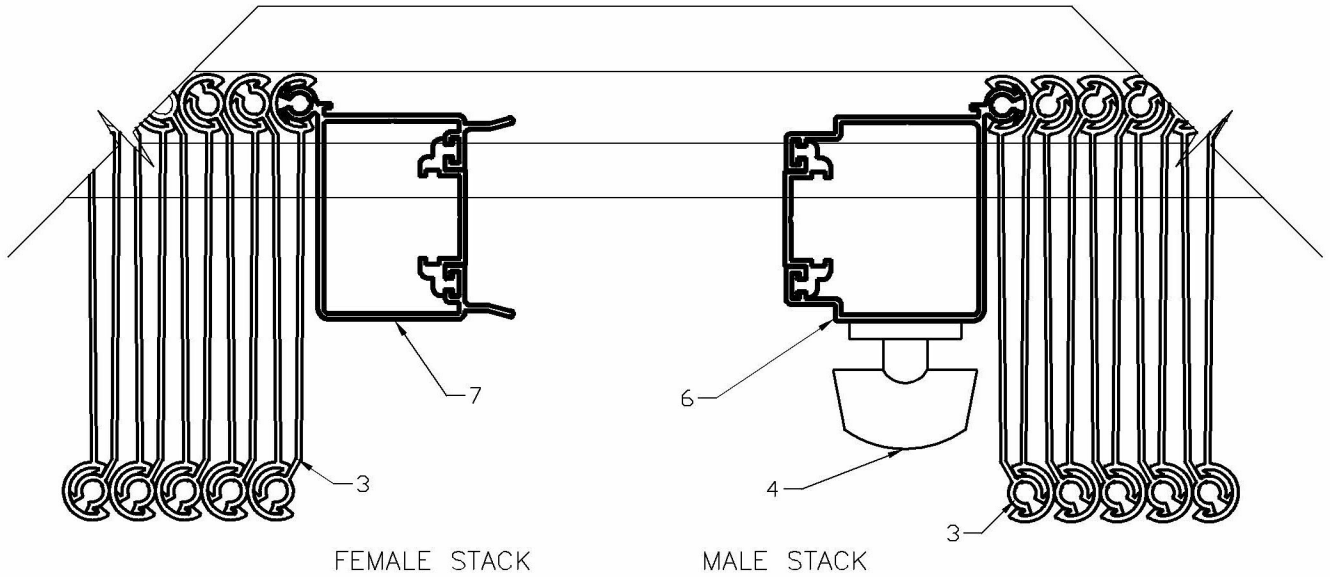
PPM Inc.

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PEDRO DE FIGUEIREDO
PE 52609

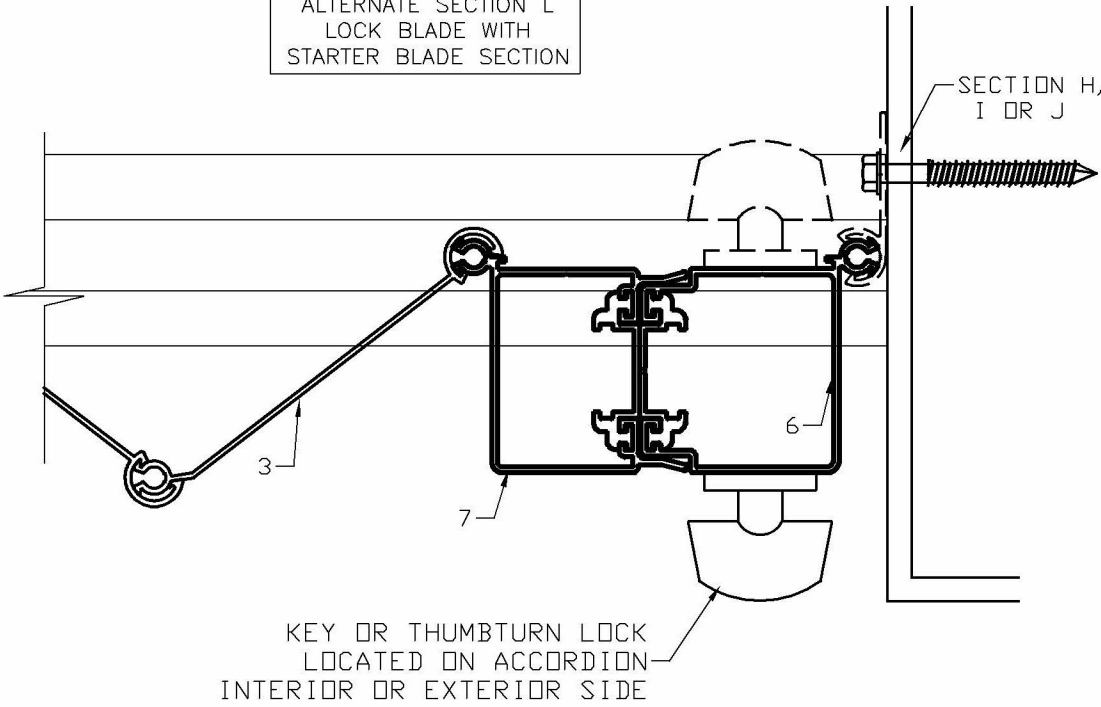
NOTE: SHUTTER CAN BE LOCKED FROM
INSIDE OR OUTSIDE BY PLACING LOCK
INSIDE OR OUTSIDE AS REQUIRED.



SECTION L
LOCK BLADE SECTION
OPEN POSITION



ALTERNATE SECTION L
ALTERNATE SECTION L
LOCK BLADE WITH
STARTER BLADE SECTION



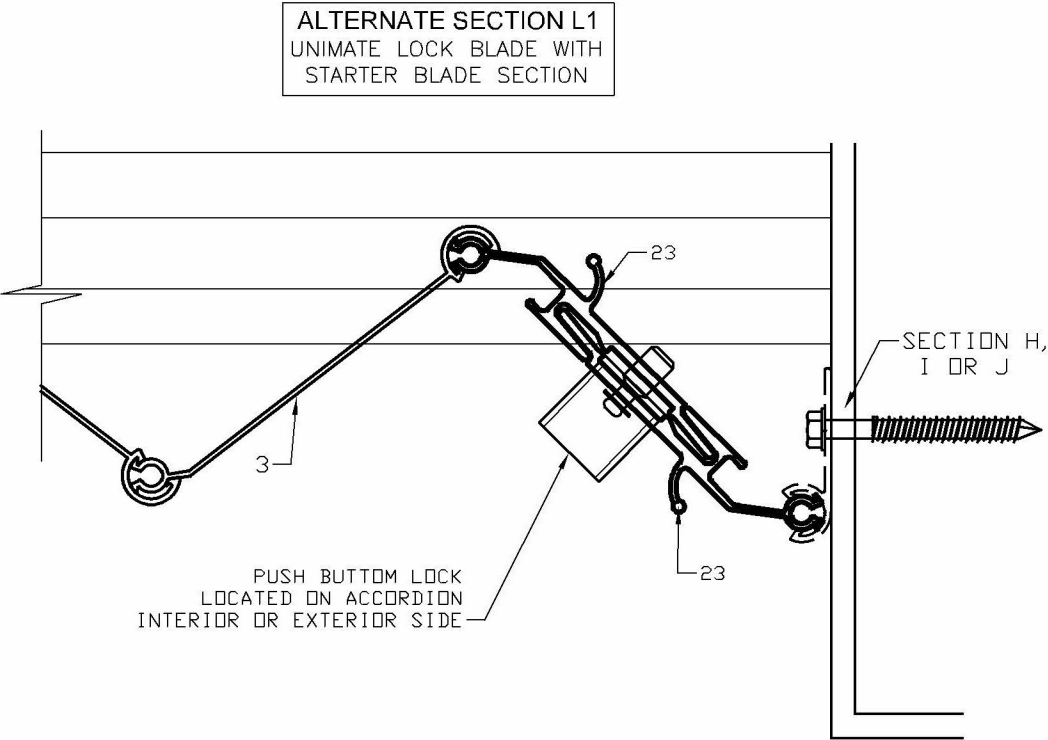
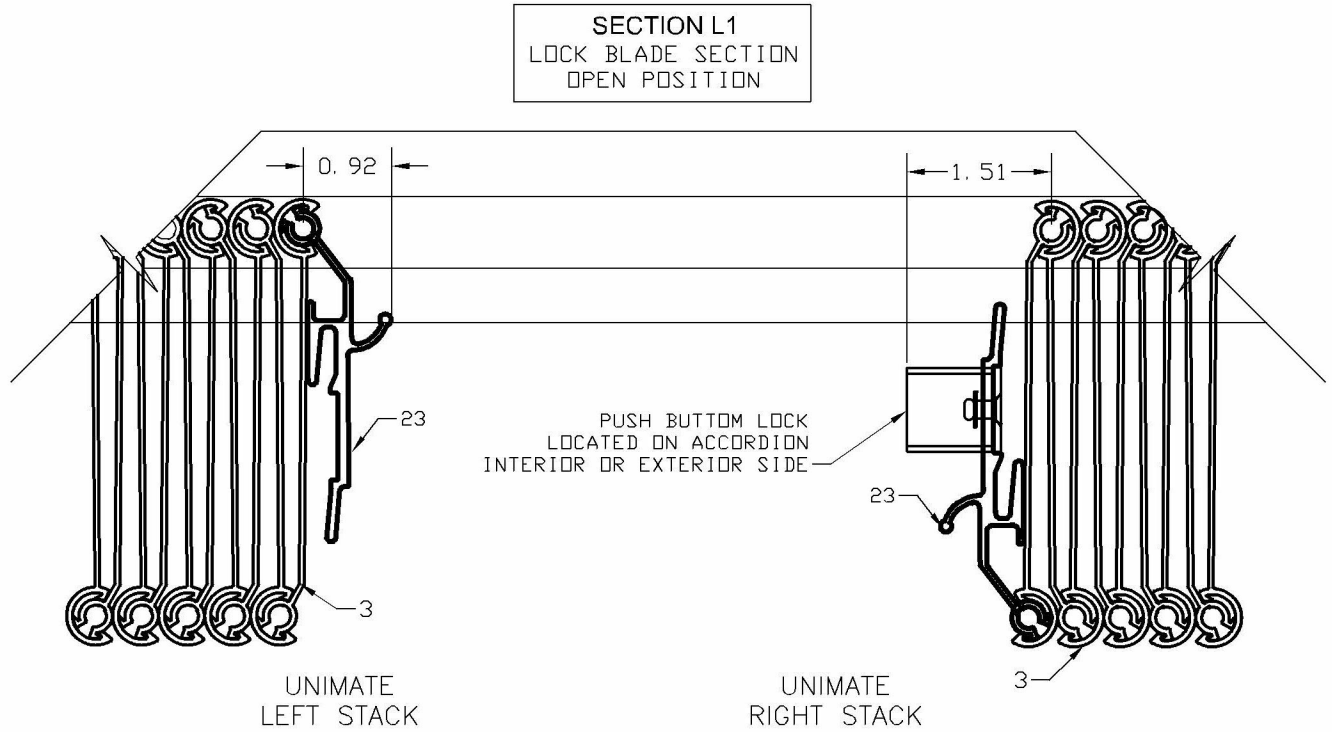
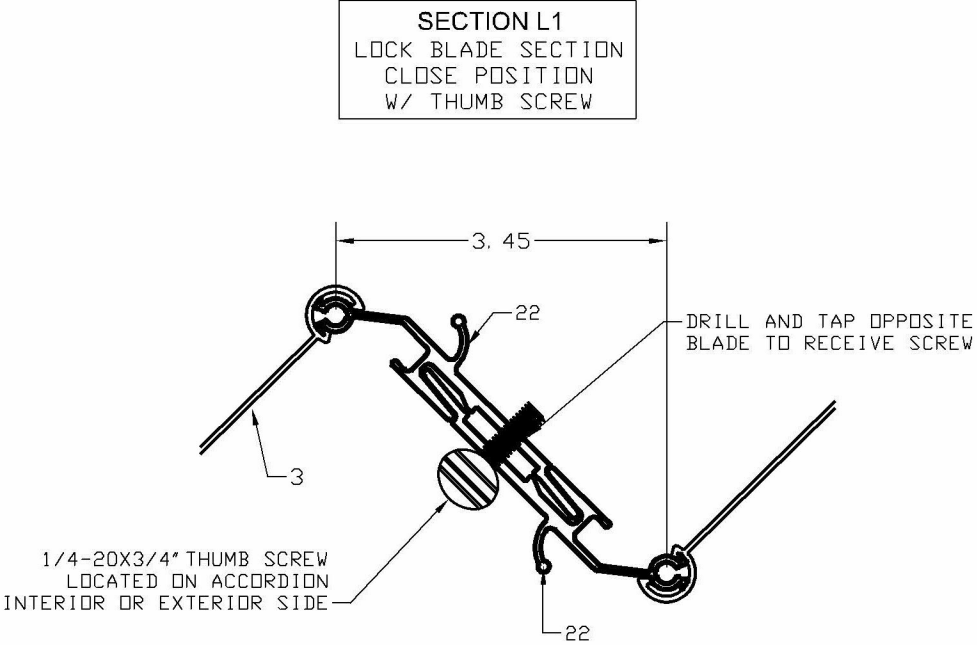
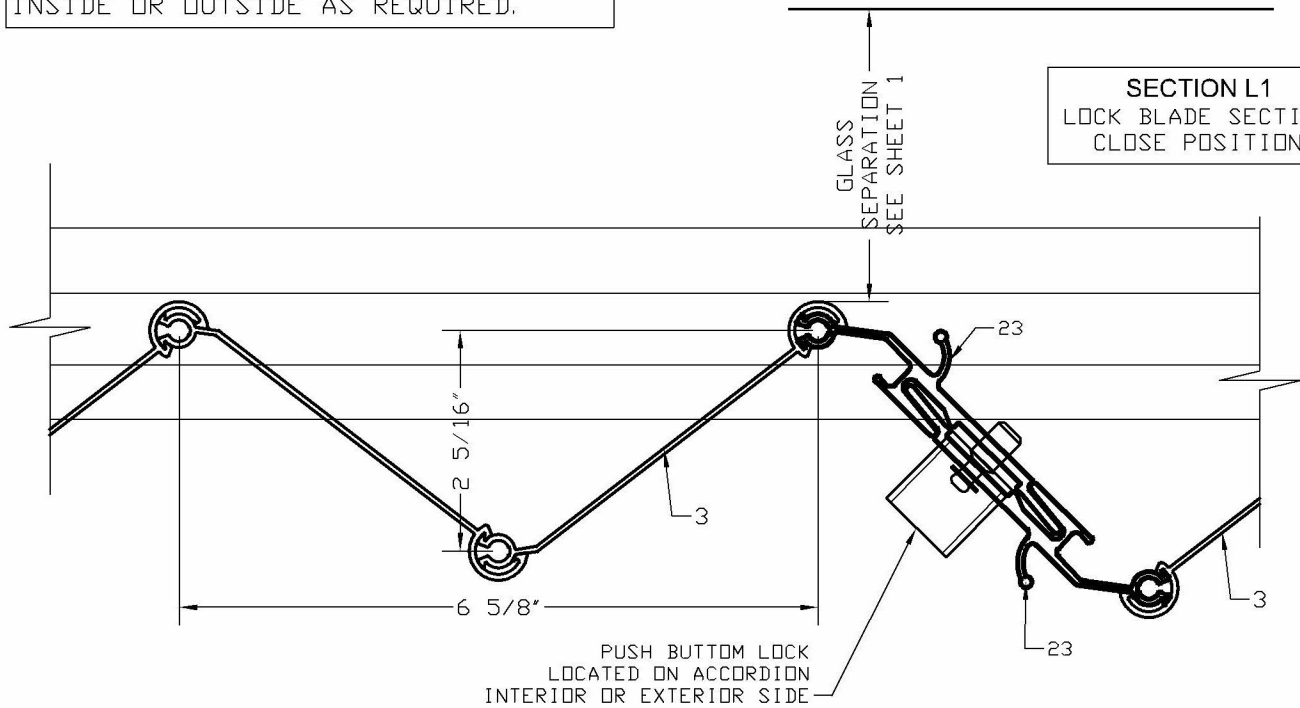
PRODUCT:
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DRAWING #	23-162
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TYPICAL HORIZONTAL
UNIMATE LOCK SECTIONS

NOTE: SHUTTER CAN BE LOCKED FROM
INSIDE OR OUTSIDE BY PLACING
THE PUSH BUTTON LOCK OR THUMB SCREW
INSIDE OR OUTSIDE AS REQUIRED.



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PE 52609
09/10/2023



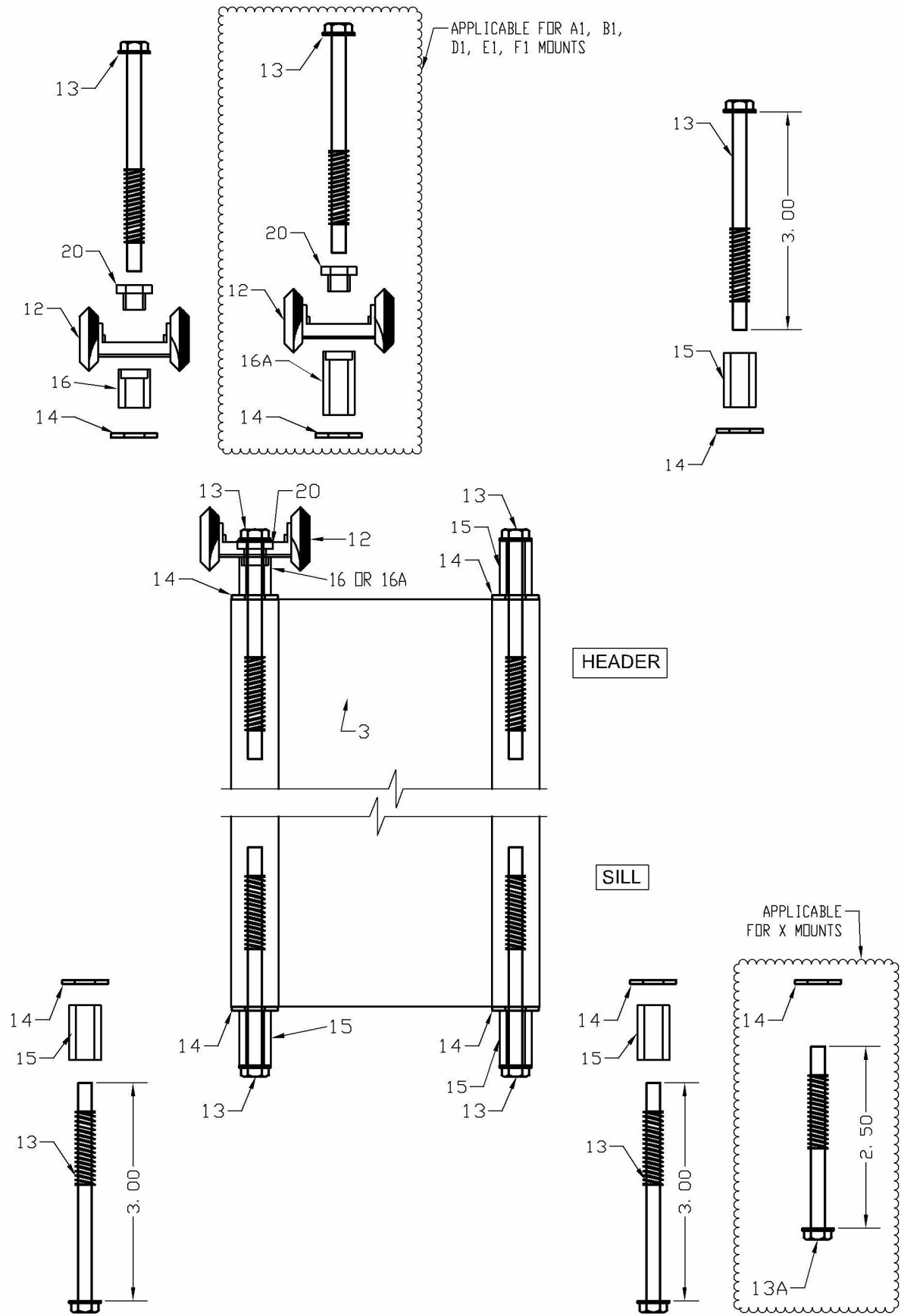
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ENGINEER SEAL
PEDRO DE FIGUEIREDO
PE 52609

TYPICAL BLADE ASSEMBLY



BILL OF MATERIALS

ITEM	DESCRIPTION	MATERIAL	MANUFACTURER
1	1" WALL MOUNT TRACK	6063-T6 ALUMINUM	CROCI
1A	2" WALL MOUNT TRACK	6063-T6 ALUMINUM	CROCI
1B	3" WALL MOUNT TRACK	6063-T6 ALUMINUM	CROCI
2	CEILING TRACK	6063-T6 ALUMINUM	CROCI
2.1	CEILING TRACK	6063-T6 ALUMINUM	CROCI
3	ACCORDION BLADE	6063-T6 ALUMINUM	CROCI
4	LOCK		
5	LOCK BLADE INSERT	6063-T6 ALUMINUM	CROCI
6	MALE LOCK BLADE	6063-T6 ALUMINUM	CROCI
7	FEMALE LOCK BLADE	6063-T6 ALUMINUM	CROCI
8	STARTER BLADE	6063-T6 ALUMINUM	CROCI
9	"A" TRACK	6063-T6 ALUMINUM	CROCI
10	ADJUSTABLE TRACK	6063-T6 ALUMINUM	CROCI
10A	ADJUST. CHANNEL	6063-T6 ALUMINUM	CROCI
10B	ADJUST. CHANNEL W/ FLANGE	6063-T6 ALUMINUM	CROCI
11			
12	WHEEL CARRIER	PLATED STEEL/ NYLON	CROCI
13	1/4"X3" SCREW BLADE	410 STAINLESS STEEL	CROCI
13A	1/4"X2.5" SCREW BLADE	410 STAINLESS STEEL	CROCI
14	1/2" SS FLAT WASHER	STAINLESS STEEL	CROCI
15	7/8"x7/16" BUSHING	NYLON	CROCI
16	1/2"x1/2" WHEEL BUSHING	NYLON	CROCI
16A	1/2"x3/4" WHEEL BUSHING	NYLON	CROCI
17	1/8 BUILT-OUT ANGLES	6063-T6 ALUMINUM	CROCI
18	1/4"-2X3 ANGLE	6063-T6 ALUMINUM	CROCI
19	PUSH-BOTTOM LOCK		
20	1/2"x 1/4" WHEEL SPACER	NYLON	CROCI
21	1" WALL MOUNT TRACK	6063-T6 ALUMINUM	CROCI
21.1	1" WALL MOUNT TRACK	6063-T6 ALUMINUM	CROCI
21A	2" WALL MOUNT TRACK	6063-T6 ALUMINUM	CROCI
21.1A	2" WALL MOUNT TRACK	6063-T6 ALUMINUM	CROCI
21B	3" WALL MOUNT TRACK	6063-T6 ALUMINUM	CROCI
21.1B	3" WALL MOUNT TRACK	6063-T6 ALUMINUM	CROCI
22	0.04" OPTIONAL COVER	ALUMINUM BREAK METAL	CROCI
23	UNIMATE LOCK BLADE	6063-T6 ALUMINUM	CROCI
24	1 1/2 X 2 1/4 RECESS TRACK	6063-T6 ALUMINUM	CROCI
24A	DUST COVER	6063-T6 ALUMINUM	CROCI
25	THRESHOLD SILL TRACK	6063-T6 ALUMINUM	CROCI
26	THRESHOLD SILL TRACK	6063-T6 ALUMINUM	CROCI
27	1" END CLOSURE	6063-T6 ALUMINUM	CROCI
28	2" END CLOSURE	6063-T6 ALUMINUM	CROCI
29	3" END CLOSURE	6063-T6 ALUMINUM	CROCI

ANCHOR SCHEDULE

ANCHOR	SUBSTRATE	DESCRIPTION	EMBED.	ED	MANUFACTURER
A	3000 PSI CONCRETE	1/4" TAPCON	1-3/4"	2-1/2	ITW
		1/4" ULTRACON	1-3/4"	2-1/2	ELCO
B	GF C90 CMU BLOCK	1/4" ULTRACON	2-1/4"	2-1/2	ELCO
		1/4" TAPCON	1-1/4"	2-1/2	ITW
C	WOOD (SG>=.55)	1/4" ULTRACON	1-1/4"	2-1/2	ELCO
		1/4" TAPCONS/ ULTRACON	1-1/2"	1.0	ITW / ELCO
D	3000 PSI CONCRETE	#14 WOOD SCREWS	1-1/2"	1.0	GENERIC
E	.125" ALUMINUM	1/4-20 SOLID-SET	7/8"	3.0	ALL POINTS
	.11" 12GA STEEL	1/4-14 TAPPING SCREWS	1/8"	3/8	GR5 GENERIC

ANCHOR NOTES:
1- ANCHOR TYPE E TO BE INSTALLED IN 6063-T6 ALUMINUM OR ASTM A36 STEEL
2- WOOD MINIMUM SPECIFIC GRAVITY SG=0.55 SOUTHERN PINE OR MIXED MAPLE
3- ANCHORS HAVE BEEN EVALUATED UNDER ALUMINUM DESIGN MANUAL 2010 AND AMERICAN WOOD COUNCIL DESIGN MANUAL NDS 2012
4- ANCHORS ARE TO BE 300 SERIES STAINLESS STEEL OR PROTECTED WITH COATING SUFFICIENT TO PROVIDE CORROSION PROTECTION FOR THE INTENDED SERVICE.

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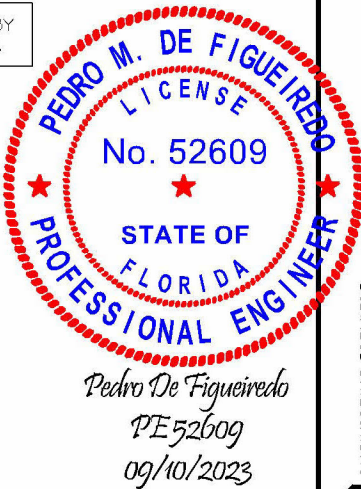
PRODUCT:
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COMPONENTS

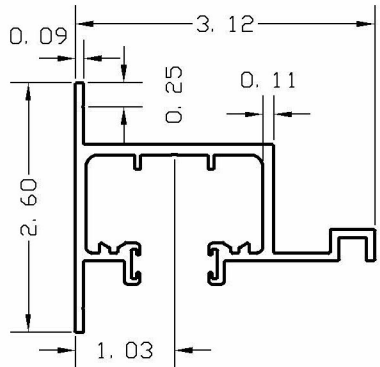
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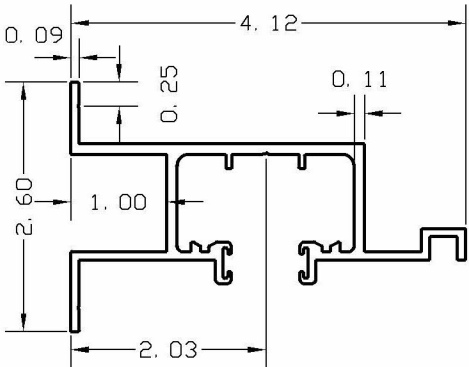
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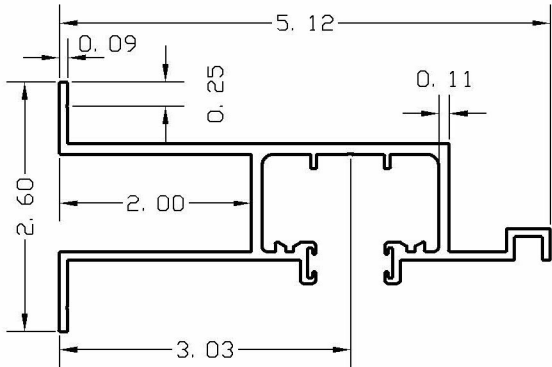
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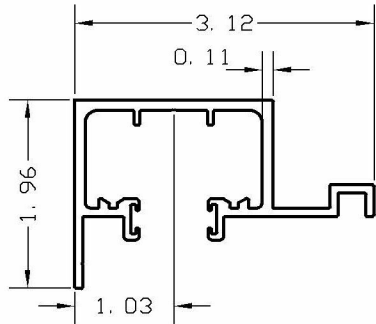
1- 1" WALL MOUNT TRACK



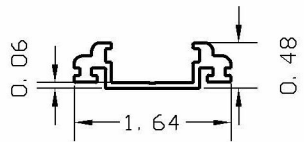
1A- 2" WALL MOUNT TRACK



1B- 3" WALL MOUNT TRACK

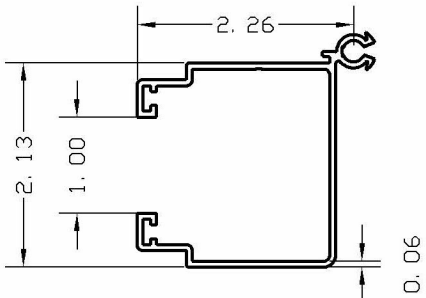


2- CEILING TRACK

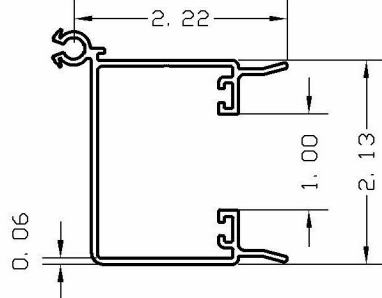


5- LOCK BLADE INSERT

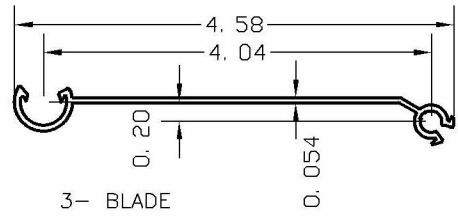
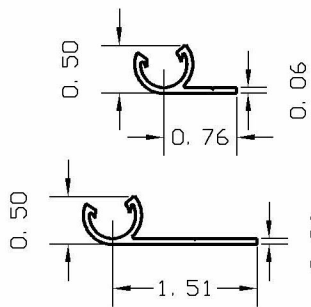
6- MALE LOCK STYLE



7- FEMALE LOCK STYLE

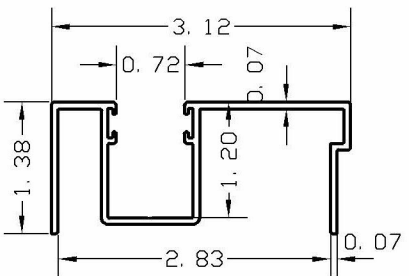
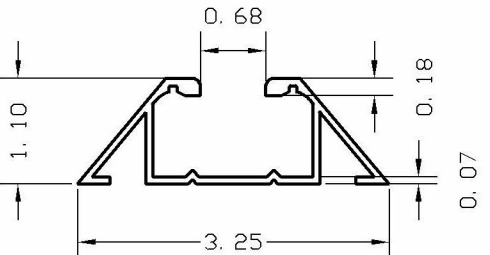


8- STARTER BLADES

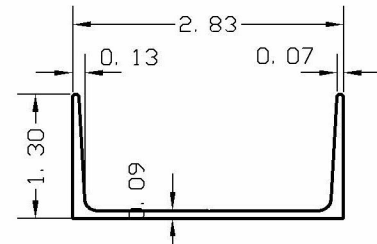


3- BLADE

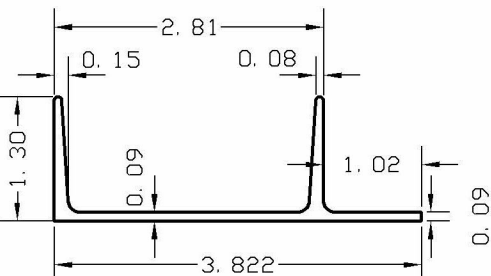
9- "A" TRACK



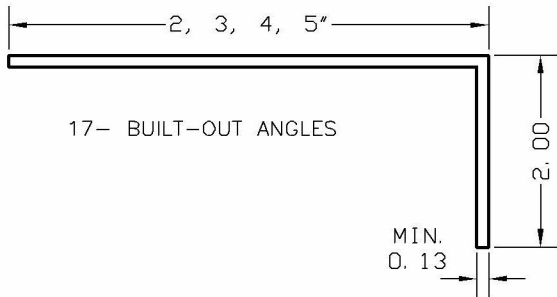
10-ADJUSTABLE SILL TRACK



10A- ADJUSTABLE SILL CHANNEL

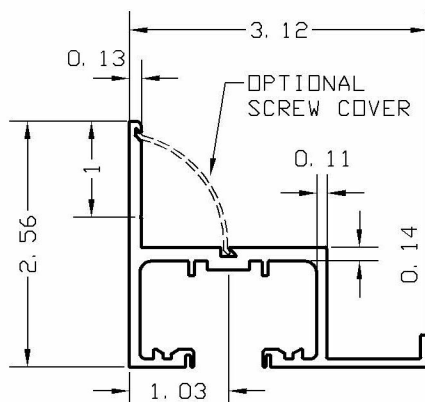
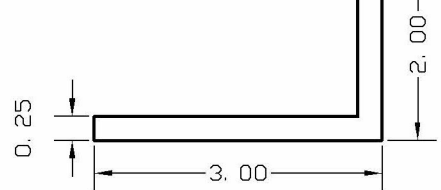


10B- ADJUSTABLE SILL CHANNEL W/ FLANGE

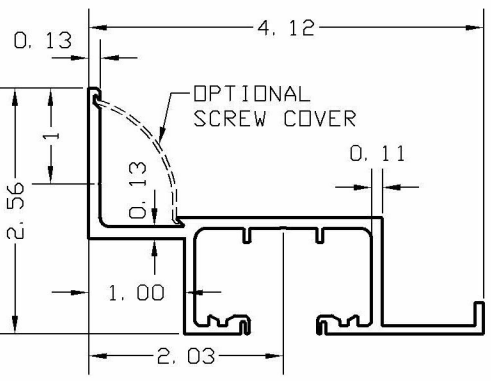


17- BUILT-OUT ANGLES

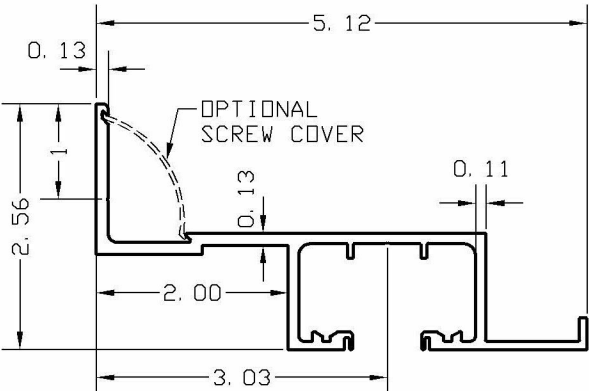
18- MOUNT TYPE E ANGLE



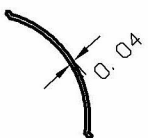
21- 1" WALL MOUNT TRACK



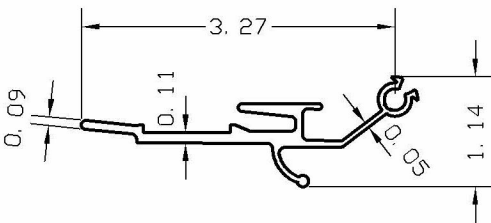
21A- 2" WALL MOUNT TRACK



21B- 3" WALL MOUNT TRACK



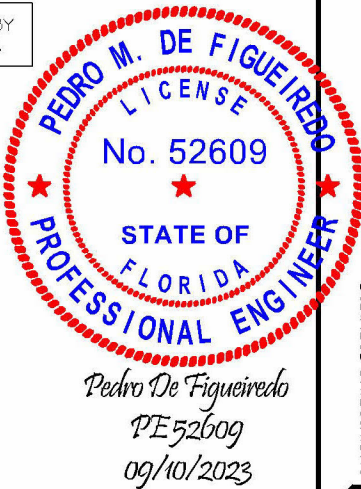
22- 0.04" BREAK METAL
OPTIONAL SCREW COVER



23- UNIMATE LOCK STYLE

COMPONENTS

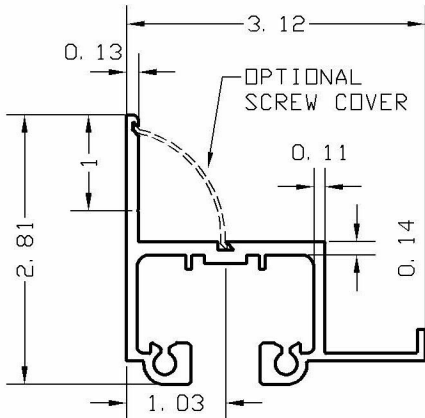
THIS ITEM HAS BEEN DIGITALLY SIGNED AND SEALED BY
PEDRO DE FIGUEIREDO ON THE DATE INDICATED HERE.



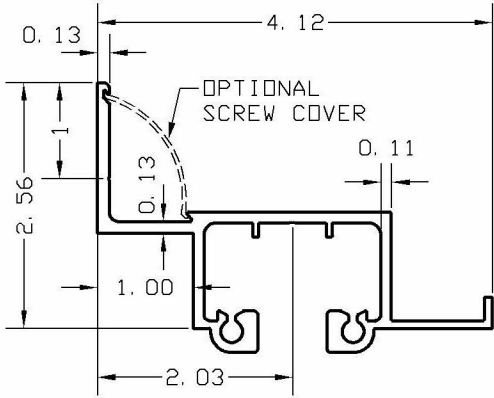
PRODUCT:
CROCI
ACCORDION SHUTTER
EXTRUDED ALUMINUM

MANUFACTURER:
CROCI
NORTH AMERICA
11600 ADELMO LANE FORT MYERS, FL 33966
TEL.: (239) 278-3066 - FAX: (239) 278-5977
EMAIL: INFO@CROCIUSA.COM

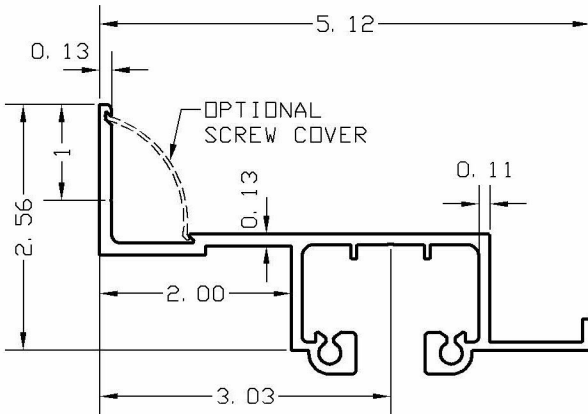
SCALE:
NA
DATE:
09/10/2023
DRAWING #
23-162
SHEET
12 of 12



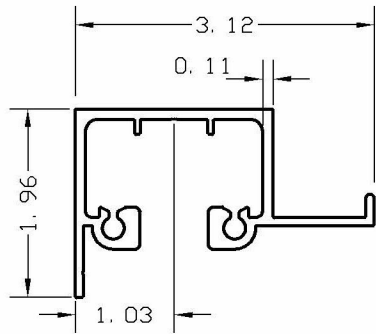
21.1- 1" WALL MOUNT TRACK



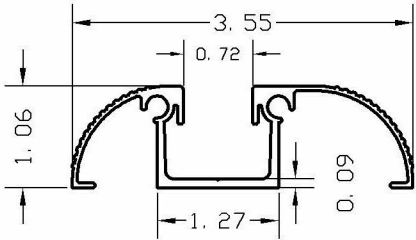
21.1A- 2" WALL MOUNT TRACK



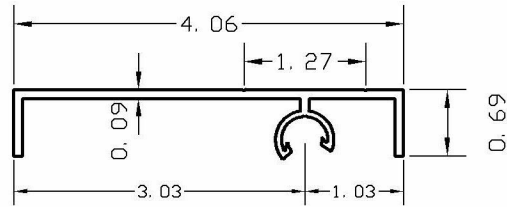
21.1B- 3" WALL MOUNT TRACK



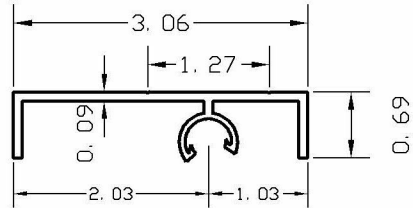
2.1- CEILING TRACK



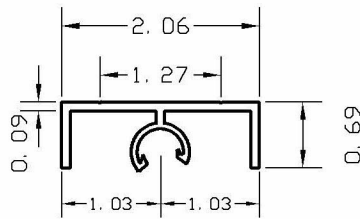
25- THRESHOLD SILL TRACK



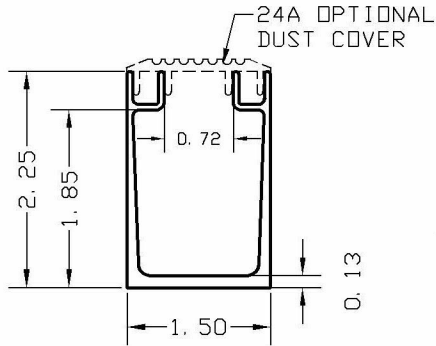
27B- 3" END CLOSURE



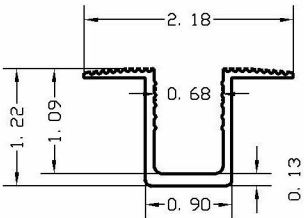
27A- 2" END CLOSURE



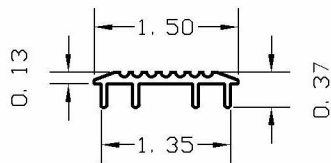
27- 1" END CLOSURE



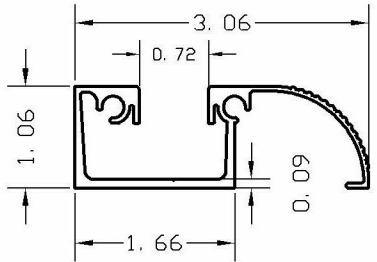
24- RECESS FLOOR TRACK



24.1- ALTERNATE
RECESS FLOOR TRACK



24A- DUST COVER



26- THRESHOLD SILL TRACK